SELECTIVE BLOCKWISE TRANSFER FOR SURROGATE-ASSISTED PREDICTION UNDER POPULATION SHIFT

Xiangyu Wang
wangxiangyu1311@gmail.com

October 2026

ABSTRACT

When a label-scarce target population borrows from a label-rich source, taking the source whole or refusing it whole gives up information if only part of it transfers. We propose a blockwise transfer rule for surrogate-assisted prediction under population shift. The surrogates enter only an imputation model for the unlabelled target sample, and the deployed predictor reads the image alone. Splitting the imputation parameter into an image block and a surrogate block leaves four candidates, and a cross-fitted target log loss with a prespecified dead zone selects among them. On the criterion the rule shares with a binary accept-or-refuse decision, its selection is never worse, and the same four losses give an exact two-block Shapley allocation.

We assess the rule in simulation and in an application to pneumothorax prediction in CheXpert-Plus, with domains defined by the recorded race field. Blockwise selection pays where the binary rule refuses the source outright. The dead zone, the one input for which the existing theory gives no computable value, can be fixed by a prespecified schedule without the decisions turning on it. The attribution orders the two blocks by transferability only where they are comparably strong.

1 Introduction

We consider prediction from medical images in a setting where labelled outcomes are scarce in both of the datasets available while unlabelled images are plentiful in both, a situation that arises whenever the label requires expert annotation and the images do not [1, 2]. Too few labels remain to fit a network end to end, and the unlabelled images go unused unless something stands in for the missing outcomes. In biomedical records such information is usually recorded already, in the form of features filed alongside the image that are predictive of the outcome without being the outcome, and a line of work has used exactly these to impute labels for unlabelled observations and estimate a risk model from them [4]. What is scarce is the confirmed outcome rather than the record: labels extracted from reports scale with the archive, but what a radiologist perceives and what the report states diverge enough that the two cannot be treated as the same quantity [3].

STRIFLE [5] carries that strategy across two populations. Methods for a single population leave the labels of a second one unused, and those are the labels most likely to be plentiful here, since a population that has been imaged longer has accumulated more confirmed outcomes. Transferring them is the subject of a literature that writes the source and target parameters as a sparse contrast and corrects it under an ℓ_1 budget [6, 7, 8]. STRIFLE combines a label-rich source, a label-scarce target, and unlabelled observations from both, estimating a density ratio for covariate-shift adjustment together with an imputation model that assigns soft labels to unlabelled target rows, with guarantees stated to survive misspecification of either nuisance model.

We apply it to pneumothorax prediction from chest radiographs in CheXpert-Plus [11, 13]. Pneumothorax is air in the pleural space and can progress to a life-threatening emergency; the radiograph is the first test ordered when it is suspected, and a small one is easy to miss on a frontal film [14]. The setting fits the method because each radiograph is filed with a report whose History sections are written before the image is read, and those sections record the workup: a request to rule out a pneumothorax says someone had already suspected one, a mention of a chest tube that the patient is already being treated for one. That is exactly the context an imputation model needs and the image does not carry,

and exactly the reason it cannot enter the deployed model, since a predictor reading it would be told the answer rather than finding it in the pixels. The cohort does not itself put us in the regime above, its labels being extracted from the reports and as plentiful as the images, so we restrict the labelled target rows to the size at which the question arises and leave the rest unlabelled. What the study varies is the regime and not the resource.

The two populations are the two largest racial groups in the cohort. That is not a neutral split for this method: the image alone carries enough of the distinction [17, 16] that a model fitted on one group can behave differently on the other [15, 18], so the transfer the method has to decide about is one that could plausibly fail. What the model reads is two things of different kinds. The image enters as a frozen representation [22, 19], the labels being too few to train a network here, and the surrogate as text terms and recorded fields. A shift in how a representation distributes its coordinates across populations and a shift in how a history is written are not the same event, and nothing makes them fail together. Both are also diffuse: a learned representation spreads the signal across its coordinates rather than concentrating it in a few.

STRIFLE decides about the two together. Its transferability check compares the corrected meta fit against a target-only one and either accepts the transfer whole or discards it whole, so a defect confined to one kind of feature costs the other. The check also carries a threshold ϵ_0 that the original treatment sets to zero under an identity link and an exactly sparse $\bar{\theta}$ [5, Sec. S.4]. Neither holds here: the outcome is binary and the coefficients of a learned representation are diffuse rather than sparse. The same section allows zero in the remaining case too, once the unlabelled samples are large enough, without saying how large. What is left is the instruction to choose a small positive value, and the condition that would justify a particular one is stated in quantities no application can evaluate.

We take up both problems, and a third that the first one raises.

1. **Selective blockwise transfer.** We split the imputation parameter into the coordinates the deployed model reads and those it does not, and replace the accept-or-discard decision with a choice among the four coalitions they generate. On the criterion the two rules share, the coalition rule is never worse, by an identity that assumes nothing. What it is worth at deployment is another matter, and turns on whether the block it retains is the one the estimand reads.
2. **An operational threshold and its assessment.** For the dead zone the original treatment leaves unspecified outside the linear case, we prespecify a schedule indexed by dimension and target sample size. Where δ_0 is known we can ask what that choice costs against the best a constant could buy, and we report it. The sufficient condition itself we do not verify, and we say by how much the schedule falls short of it.
3. **Attribution by block, and its limits.** Deciding by block invites the question of which block is at fault, and the same four losses answer it exactly, as a two-block Shapley allocation [9]. Reading that allocation as a measure of transferability requires the two blocks to be comparably strong, and it speaks about the layer it is defined on rather than the one that is deployed.

Section 2 develops the estimator: the two blocks and the four coalitions they generate, the criterion the two rules share, the threshold schedule, and the allocation. Section 3 has two halves. The first varies one thing at a time, sweeping where the defect sits and how sparse it is, how strong the two blocks are relative to one another, and how much labelled target data there is, over $4{,}350$ replicates in which δ_0 is known exactly rather than approximated. The second adds no design and puts the three claims above to those replicates: what the coalition rule buys over the binary one, what the threshold costs against the best a constant could buy, and what the allocation measures.

Section 4 carries the same procedure to pneumothorax prediction under three image representations and two labelled-target budgets, where δ_0 is unobservable and the strength condition can be checked but not the transfer itself. Only one branch at one budget meets that condition, and we read the allocation there and nowhere else. Section 5 states what the evidence does not cover.

2 Methodology

2.1 Setup and notation

The setup and notation of this section follow Cai et al. [5] except where stated; Section 2.2 lists what the construction inherits and where it departs.

Let $Y \in \{0, 1\}$ denote the binary clinical endpoint, $X \in \mathbb{R}^p$ a vector of predictors derived from the diagnostic image, and $S \in \mathbb{R}^q$ a vector of surrogate features derived from the accompanying record. Write $Z = (X^\top, S^\top)^\top \in \mathbb{R}^d$ with $d = p + q$, and let $R \in \{0, 1\}$ indicate the population, $R = 0$ for the label-rich source $\mathcal{S}$ and $R = 1$ for the label-scarce target $\mathcal{T}$. Within each population we distinguish a labelled and an unlabelled stratum. Let $I_{\iota,\jmath}$ denote the index set

of observations from population $\iota \in \{\mathcal{S}, \mathcal{T}\}$ and stratum $\jmath \in \{L, U\}$, with cardinalities $n_{\mathcal{S}} = |I_{\mathcal{S},L}|$, $N_{\mathcal{S}} = |I_{\mathcal{S},U}|$, $n_T = |I_{\mathcal{T},L}|$, $N_T = |I_{\mathcal{T},U}|$, $n = n_{\mathcal{S}} + n_T$ and $N = N_{\mathcal{S}} + N_T$. We abbreviate the four datasets used in estimation as SL, SU, TL and TU, and write E_ι for expectation under population ι, $\hat{E}_\iota$ for its sample analogue, and $p_\cdot^{(\iota)}$ for the corresponding densities.

The outcome is observed only on the labelled strata; Z is observed everywhere. The labelled target sample is repeatedly split (Section 4.1) into a fitting set $I_{\mathcal{T},L}$ of size n_T and a disjoint held-out set. Throughout, n_T refers to the fitting set, and no tuning step reads the held-out set; Section 2.9 states the evaluation principle that governs it.

Estimand. The target of estimation is the outcome model in the target population. Let $g(\cdot)$ be the inverse logit. The parameters $(\alpha_0, \beta_0) \in \mathbb{R} \times \mathbb{R}^p$ solve

$$E_{\mathcal{T}}\big[(1, X^\top)^\top \big\{Y - g(\alpha + X^\top \beta)\big\}\big] = 0, \tag{1}$$

and we refer to β_0 alone where the intercept plays no role. Two features of this definition carry through the rest of the paper. First, (1) defines β_0 as the solution of an estimating equation rather than as the truth of a generative model. It is the score of a concave quasi-likelihood, strictly concave wherever $E_{\mathcal{T}}[(1, X^\top)^\top (1, X^\top)]$ is non-singular, which the whitening of Section 4.2 delivers by construction, so β_0 is well defined whether or not the logistic form describes $E_{\mathcal{T}}(Y \mid X)$. Second, it does not ask β_0 to be sparse, and the image block is a dense whitened representation. Because β_0 is estimated on $n_T + N_T$ rows while the imputation parameter of Section 2.4 must be corrected on the n_T labelled target rows alone, it may be considerably the denser of the two; that asymmetry is what makes the semi-supervised stage worth running [5, Remark 4].

The surrogates do not enter (1)**.** The deployed model reads the image alone, and S reaches β_0 only through the imputed labels of Section 2.8. In Cai et al. [5] this restriction is temporal, their surrogates being recorded no earlier than the outcome they proxy. Ours are not, and we impose it for a different reason. The surrogates are informative about Y partly through references to the finding and to its management (Section 4.1), so a model reading Z would answer a question about the image and the record together, and its performance would not be attributable to the image. The gain from S is therefore routed through the imputation model rather than through β.

Covariate shift. The method is derived under the covariate-shift structure of Cai et al. [5]: the two populations share the conditional law of the outcome given the full feature vector,

$$p^{(\mathcal{S})}_{Y|Z}(y \mid z) = p^{(\mathcal{T})}_{Y|Z}(y \mid z), \tag{2}$$

while $p^{(\iota)}_Z$ and, in particular, $p^{(\iota)}_{S|X}$ may differ, so that $p^{(\mathcal{S})}_{Y|X} \neq p^{(\mathcal{T})}_{Y|X}$ in general. That (2) holds on Z and not on X is what the surrogates are for: S has to carry the clinical context that makes the outcome law comparable across the two populations. It is what lets labelled source rows inform the imputation model of Section 2.4, and is assumed for nothing else, since by (1) the outcome model never has to transfer. A departure from it reaches the estimator through one channel, the transferability defect δ_0 of Section 2.4.

Blocks. Z partitions into the image block X and the surrogate block S, and we write $\theta_{\mathsf{X}} \in \mathbb{R}^p$ and $\theta_{\mathsf{S}} \in \mathbb{R}^q$ for the corresponding coefficient blocks of the imputation model defined in Section 2.4. This is the partition the estimand already distinguishes: by (1) the image predictors have a counterpart in β and the surrogates do not, and no other split of Z has that property. Section 2.5 forms the coalitions it supports and Section 2.7 the attribution; the reasons for keeping the partition to two blocks, and for carrying the intercept outside both, are given there.

Conventions. Calligraphic $\mathcal{S}$ and $\mathcal{T}$ denote the two populations, with the target subscript abbreviated to T in the sample sizes n_T and N_T. Upright sans-serif X and S denote the two feature blocks; they also index the coefficient blocks θ_{X}, θ_{S} and the coalitions of Section 2.5. For $v \in \mathbb{R}^m$, $\|v\|_r = \sum_j |v_j|^r$ for $r \in [0, 1)$, with the convention $0^0 = 0$, so that $\|v\|_0$ is the cardinality of the support. For $r \geq 1$, and in particular for $\|\cdot\|_1$ and $\|\cdot\|_2$, the usual ℓ_r norm is meant, and $\|v\|_\infty = \max_j |v_j|$. We write $a_n \lesssim b_n$ for $a_n \leq C b_n$, and $a_n \asymp b_n$ for $C \leq a_n / b_n \leq C'$, with $C, C' > 0$ constants. The log-quasi-likelihood contribution at linear predictor u is $Q\{g(u), y\} = yu - G(u)$ with $G(u) = \log(1 + e^u)$, so that $G' = g$; we add $\ell_{\log}(y, u) = -Q\{g(u), y\}$ where the loss is needed pointwise. Penalties act on slopes only, intercepts being left unpenalised throughout, which departs from the original construction for the reason given in Section 2.5.

2.2 Overview

The procedure has four stages.

First, the unlabelled observations from both populations, SU and TU, are used to fit a working density ratio $\hat{w}(z)$ for $p_Z^{(\mathcal{T})}/p_Z^{(\mathcal{S})}$ (Section 2.3). No outcome is read here, which is what allows the ratio to be estimated on a sample far larger than the labelled one. Second, two estimators of the imputation parameter θ are formed (Section 2.4): a target-only estimator $\hat{\theta}^{(\mathcal{T})}$ fitted on TL alone, and a meta-learning estimator $\hat{\theta}^{(\mathrm{meta})}$ obtained by fitting the pooled labelled data with the source rows reweighted by $\hat{w}$ and then correcting the result on TL. Third, the two are compared on held-out target rows under a cross-fitted split of TL, and that comparison decides how much of the source-borrowing fit is retained (Section 2.5). Fourth, the retained imputation model assigns soft labels to TU, and the outcome model $\hat{\beta}$ is fitted on those imputed rows together with the observed labels on TL (Section 2.8). Only $\hat{\beta}$ is deployed. The source thus reaches $\hat{\beta}$ by way of θ and no other route, so every section from Section 2.3 to Section 2.7 concerns the estimation of θ and Section 2.8 is short.

Borrowing is not guaranteed to help. Reweighting removes the part of the discrepancy the density ratio can express; what remains is a transferability defect, defined in Section 2.4, which the correction step may fail to remove when it is large or diffuse relative to what n_T can resolve. The third stage exists for that failure mode, and its output is a decision about how much of the source to keep rather than an estimate.

What is inherited and what is new. The construction follows Cai et al. [5] in the following respects: the estimand in (1) and the covariate-shift structure in (2); the decomposition of the estimation of θ into a density-ratio model, a target-only fit, and a density-ratio-embedded meta-learning fit; the two-step form of the meta-learning estimator, in which a reweighted pooled fit is followed by a bias-correction step on TL; the hard-thresholding rule applied to the density-ratio coefficients; and a cross-fitted transferability decision guarded by a non-negative dead zone ϵ_0.

Our extension and implementation differ from that construction in four respects.

1. The original transferability decision is a single binary indicator applied to the source contribution as a whole. We instead resolve the imputation parameter along $(\theta_{\mathsf{X}}, \theta_{\mathsf{S}})$, replacing the all-or-nothing comparison by the four coalitions $\{\varnothing, \mathsf{X}, \mathsf{S}, \mathsf{XS}\}$. The same four cross-fitted losses support both selective blockwise borrowing and an exact two-block attribution.
2. The output selected by the binary rule is always feasible for the blockwise rule: when the binary rule accepts source borrowing, XS belongs to the admitted set, and when it rejects, $\varnothing$ remains available. The blockwise rule therefore cannot have larger cross-fitted selection loss than the candidate actually selected by the binary rule. This comparison concerns the shared selection criterion only; it does not imply dominance over $\min\{L(\varnothing), L(\mathsf{XS})\}$ or over independently held-out predictive performance.
3. For our logistic imputation model, the exact-sparse identity-link linear simplification under which Cai et al. [5] allow $\epsilon_0 = 0$ is unavailable. Under the remaining sparsity and regularity regimes considered in their theory, the sufficient condition for ϵ_0 depends on the unknown target projection $\bar{\theta}$, the estimation error $\hat{\theta}^{(\mathrm{meta})} - \bar{\theta}$, and restricted-strong-convexity constants, and therefore does not yield a data-computable cutoff. We prespecify
$$\epsilon_0 = \frac{\log d}{n_T}$$
as a reproducible dimension- and sample-size-indexed operational schedule. We do not claim that this choice satisfies the original sufficient condition. Section 3.6 assesses its finite-sample operating consequences and sensitivity to its multiplicative scale.
4. The density ratio is estimated through its classification representation, an implementation allowed by Cai et al. [5, Remark 1] but not used in their numerical studies; Section 2.3 gives the resulting diagnostics.

One caution belongs here rather than later. Exactness of the two-block attribution means that all four coalitions are evaluated and no Monte Carlo approximation is required. It does not by itself make the resulting ordering a block-specific measure of transferability: each marginal value also reflects the block's baseline predictive contribution. In addition, the attribution is defined using loss at the imputation layer, whereas the endpoints of Table 1 are evaluated at the deployed layer. Nothing in the construction requires these two layers to agree; Sections 2.7, 3.3, and 3.7 characterise and measure these limitations.

Algorithm 1 STRIFLE with Shapley block attribution

Require: SL, SU, TL, TU with $Z = (X^\top, S^\top)^\top$, $X \in \mathbb{R}^p$, $S \in \mathbb{R}^q$, $d = p + q$, $n_T = |\mathrm{TL}|$. Write $\eta(z;\theta) = \theta_0 + x^\top\theta_{\mathsf{X}} + s^\top\theta_{\mathsf{S}}$; all penalties act on slopes only

Ensure: $(\hat\alpha, \hat\beta)$; $(\phi_{\mathsf{X}}, \phi_{\mathsf{S}})$; $\hat\rho$; $\hat c$

1: $\hat w \leftarrow$ thresholded density ratio fitted on SU $\cup$ TU, evaluated on SL (Sec. 2.3)

Cross-fitted coalition losses

2: Split TL into stratified folds F_1, F_2

3: **for** $k = 1, 2$ **do**

4: $(\theta^{(\mathcal{T})}, \theta^{(\mathrm{meta})}) \leftarrow$ FITPAIR$(F_k; \mathrm{SL}, \hat w)$ ▷ one pair per fold, shared by all four coalitions

5: **for all** $c \subseteq \{\mathsf{X}, \mathsf{S}\}$ **do**

6: $\theta_b \leftarrow \theta_b^{(\mathrm{meta})}$ if $b \in c$, else $\theta_b^{(\mathcal{T})}$, for $b \in \{\mathsf{X}, \mathsf{S}\}$

7: $\theta_0 \leftarrow$ unpenalised intercept refit on F_k at fixed $(\theta_{\mathsf{X}}, \theta_{\mathsf{S}})$

8: $L_k(c) \leftarrow |F_{3-k}|^{-1} \sum_{i \in F_{3-k}} \ell_{\log}\{Y_i, \eta(Z_i;\theta)\}$

9: **end for**

10: **end for**

11: $L(c) \leftarrow \frac{1}{2}\{L_1(c) + L_2(c)\}, \quad v(c) \leftarrow L(\varnothing) - L(c)$

Attribution and decisions

12: $\phi_{\mathsf{X}} \leftarrow \frac{1}{2}\big[\{v(\mathsf{X}) - v(\varnothing)\} + \{v(\mathsf{XS}) - v(\mathsf{S})\}\big], \quad \phi_{\mathsf{S}} \leftarrow \frac{1}{2}\big[\{v(\mathsf{S}) - v(\varnothing)\} + \{v(\mathsf{XS}) - v(\mathsf{X})\}\big]$

13: $\epsilon_0 \leftarrow \log d / n_T, \quad \mathcal{E} \leftarrow \{\varnothing\} \cup \{c : L(c) \le L(\varnothing) - \epsilon_0\}$

14: $\hat\rho \leftarrow \mathbf{1}\{L(\mathsf{XS}) \le L(\varnothing) - \epsilon_0\}$ ▷ binary rule, Cai et al. [5, eq. 9]

15: $\hat c \leftarrow \arg\min_{c \in \mathcal{E}} L(c)$ ▷ coalition rule

Deployment

16: $(\theta^{(\mathcal{T})}, \theta^{(\mathrm{meta})}) \leftarrow$ FITPAIR(TL; SL, $\hat w$); form $\theta^{(\hat c)}$ as in line 6 and refit its intercept on TL

17: $m_i \leftarrow g\{\eta(Z_i; \theta^{(\hat c)})\}$ for $i \in$ TU

18: $(\hat\alpha, \hat\beta) \leftarrow \arg\min_{\alpha,\beta} -\frac{1}{N_T + n_T}\big[\sum_{i \in \mathrm{TU}} Q\{g(\alpha + X_i^\top\beta), m_i\} + \sum_{i \in \mathrm{TL}} Q\{g(\alpha + X_i^\top\beta), Y_i\}\big] + \lambda_\beta\|\beta\|_1$

19: **procedure** FITPAIR(A; SL, $\hat w$)

20: $\theta^{(\mathcal{T})} \leftarrow \arg\min_\theta -|A|^{-1} \sum_{i \in A} Q[g\{\eta(Z_i;\theta)\}, Y_i] + \lambda_{\mathcal{T}}\|\theta\|_1$

21: $\tilde\theta \leftarrow \arg\min_\theta -\frac{1}{|\mathrm{SL}|+|A|} \sum_{i \in \mathrm{SL}\cup A} \hat w_i^{\mathbf{1}(i \in \mathrm{SL})} Q[g\{\eta(Z_i;\theta)\}, Y_i] + \lambda_\theta\|\theta\|_1$

22: $\hat\delta \leftarrow \arg\min_\delta -|A|^{-1} \sum_{i \in A} Q[g\{\eta(Z_i;\tilde\theta) + \eta(Z_i;\delta)\}, Y_i] + \lambda_\delta\|\delta\|_1$

23: **return** $\theta^{(\mathcal{T})}$ and $\theta^{(\mathrm{meta})} = \tilde\theta + \hat\delta$ ▷ coordinatewise, intercept included

24: **end procedure**

2.3 Density ratio model

The first nuisance model is the density ratio $w(z) = p_Z^{(\mathcal{T})}(z)/p_Z^{(\mathcal{S})}(z)$, estimated on the unlabelled rows of both populations. We estimate it by binary classification, the route sanctioned in Cai et al. [5, Remark 1 and Supplement S.1], which carries the assumption $E_{\mathcal{S}}[\{\hat w(Z) - w_0(Z)\}^2] = o(1)$ with $\sup_z w_0(z)/\hat w(z)$ bounded in probability. Let $R_i = 1$ for rows drawn from TU and $R_i = 0$ for rows drawn from SU, and fit a logistic classifier on the pooled unlabelled rows of Z (the full feature vector whose densities w compares) under a fixed ridge penalty, which stabilises the optimisation and selects nothing. Writing $\hat\eta(z)$ for the fitted logit,

$$\log \hat w(z) \;=\; \hat\eta(z) \;+\; \log\left(\frac{N_{\mathcal{S}}}{N_T}\right), \tag{3}$$

with $N_{\mathcal{S}} = |I_{\mathcal{S},U}|$ and $N_T = |I_{\mathcal{T},U}|$, the offset converting posterior odds of domain membership into a density ratio [10]. The classifier is fitted on SU $\cup$ TU and applied to SL, the rows that carry $\hat w$ in (8), which are disjoint from the rows it was fitted on.

Thresholding. The design matrix is standardised on the pooled unlabelled rows, the hard threshold $\kappa = 10\sqrt{\log d/(N_{\mathcal{S}} + N_T)}$ of Cai et al. [5] is applied to the standardised coefficients, and the survivors are mapped back to the original scale. That cutoff is defined for covariates of unit variance; applied on any other scale it would reweight the pruning by column variance.

Centring, clipping and renormalisation. The log ratio is centred so that $E_{\mathcal{S}}[\hat w] = 1$, the centred weights are clipped, and the result is renormalised. Centring is invariant to an additive constant in the log ratio, so neither the fitted intercept

nor the offset in (3) reaches $\hat{w}$. One consequence is worth recording: $\hat{w}$ does not depend on N_S/N_T, a design quantity rather than a population one, so the weights are unchanged by how much unlabelled data is drawn from each population. We retain both terms because (3) is the density ratio and the classifier's logit is not. The order of the three operations is fixed by that same invariance: clipping is not shift-equivariant, so clipping first would let a systematic offset be absorbed asymmetrically at the bounds, and renormalising afterwards could not undo it. Clipping is an addition to Cai et al. [5], adopted to enforce in finite samples the bounded $\sup_z w_0(z)/\hat{w}(z)$ that the assumption above requires.

Conditions monitored. Two quantities decide whether the fitted weights may be read as a density ratio at all, and both are reported rather than assumed. First, a domain classification that separates the two populations almost perfectly leaves w barely identifiable and the bounded-ratio condition doubtful; we prespecify a separability threshold beyond which a fit's transfer results are not read without qualification, and treat this as a precondition for interpreting them rather than as a summary of them. Separability is measured by the apparent domain-classification AUC, which is not below its out-of-sample counterpart, so a fit clearing the threshold on the apparent value clears it a fortiori and the check can only over-flag. Second, the classification route constrains no moment of the reweighted source design, and centring restores only the mean, whereas the calibration-loss estimator of Cai et al. [5] balances every coordinate by construction; we therefore report the realised imbalance

$$\mathcal{B}_\infty = \left\| \hat{E}_{\mathcal{S}}[\hat{w}Z] - \hat{E}_{\mathcal{T}}[Z] \right\|_\infty, \tag{4}$$

the empirical counterpart of the identity $E_{\mathcal{S}}[w_0(Z)h(Z)] = E_{\mathcal{T}}[h(Z)]$ at $h(z) = z$ that the classification route leaves unenforced. Effective sample size, clipped fraction and retained support are recorded alongside them.

2.4 Imputation model: target-only and meta-learning estimators

The second nuisance model is the imputation model $m(z) = E(Y \mid Z = z) = g\{\eta(z;\theta)\}$, common to both populations under (2). We write

$$\eta(z;\theta) = \theta_0 + x^\top \theta_{\mathsf{X}} + s^\top \theta_{\mathsf{S}},$$

so that $\theta = (\theta_0, \theta_{\mathsf{X}}^\top, \theta_{\mathsf{S}}^\top)^\top$ carries an intercept together with one coefficient block per component of Z: $\theta_{\mathsf{X}} \in \mathbb{R}^p$ on the image predictors and $\theta_{\mathsf{S}} \in \mathbb{R}^q$ on the surrogates. Neither X nor S contains a constant; the intercept is carried explicitly and belongs to neither block.

Let $\bar{\theta} = \arg\max_\theta E_{\mathcal{T}}[Q\{g(\eta(Z;\theta)), Y\}]$ be the population maximiser in the target distribution. Its first-order conditions make the residual $Y - g\{\eta(Z;\bar{\theta})\}$ orthogonal to the constant and to every coordinate of Z, and X sits inside Z, so

$$E_{\mathcal{T}}\left[(1, X^\top)^\top \{Y - \bar{m}(Z)\}\right] = 0, \qquad \bar{m}(z) = g\{\eta(z;\bar{\theta})\}. \tag{5}$$

This is a property of the projection, not of the model being right, so (5) survives even when $g\{\eta(z;\theta)\}$ describes $E(Y \mid Z)$ badly. It is what keeps (α_0, β_0) unbiased under a crude imputation model in Section 2.8. Note that (5) matches (1) coordinate for coordinate, intercept included.

Target-only estimator. Using TL alone,

$$\hat{\theta}^{(\mathcal{T})} = \arg\min_\theta \left\{ -\frac{1}{n_T} \sum_{i \in I_{\mathcal{T},L}} Q[g\{\eta(z_i;\theta)\}, y_i] + \lambda_{\mathcal{T}} \|\theta\|_1 \right\}. \tag{6}$$

The ℓ_1 penalty [23] acts on slopes only here and throughout. It touches no source data, which is what makes it the reference point for everything below.

Weighted meta-learning with bias correction. The source enters through a second population parameter. Writing w_0 for the limit of the working density ratio and $a = n_{\mathcal{S}}/n$,

$$\theta^* = \arg\max_\theta \left\{ a\, E_{\mathcal{S}}[w_0(Z)\, Q[g\{\eta(Z;\theta)\}, Y]] + (1-a)\, E_{\mathcal{T}}[Q[g\{\eta(Z;\theta)\}, Y]] \right\}, \qquad \delta_0 = \bar{\theta} - \theta^*, \tag{7}$$

following Cai et al. [5, eq. 6]. The defect δ_0 is the object promised in Section 2.1: under (2) it vanishes whenever either nuisance model is correctly specified, because a correct w makes the tilted source term an expectation under $\mathcal{T}$, and a correct m makes both terms maximised at the same θ. It is therefore the single channel through which misspecification of both nuisance models, or a departure from (2), reaches the procedure.

We estimate θ^* on SL $\cup$ TL,

$$\tilde{\theta} = \arg\min_\theta \left\{ -\frac{1}{n} \sum_{i \in I_{\mathcal{S},L} \cup I_{\mathcal{T},L}} \hat{w}(z_i)^{\mathbf{1}(i \in I_{\mathcal{S},L})} Q[g\{\eta(z_i;\theta)\}, y_i] + \lambda_\theta \|\theta\|_1 \right\}, \tag{8}$$

so source rows carry the estimated density ratio and target rows carry unit weight, and then estimate δ_0 on the target labelled rows, holding $\tilde{\theta}$ fixed as an offset:

$$\hat{\delta} = \arg\min_{\delta} \left\{ -\frac{1}{n_T} \sum_{i \in I_{\mathcal{T},L}} Q\Big[g\{\eta(z_i;\tilde{\theta}) + \eta(z_i;\delta)\}, y_i \Big] + \lambda_\delta \|\delta\|_1 \right\}, \qquad \hat{\theta}^{(\mathrm{meta})} = \tilde{\theta} + \hat{\delta}. \tag{9}$$

That the penalty acts on slopes only matters here in particular: a difference in prevalence between the two populations is a pure calibration offset, and there is no reason to make the correction buy it out of the ℓ_1 budget before it can touch a single slope.

What the correction can and cannot recover. Problem (9) is convex, but whether it recovers δ_0 rather than merely shrinking it turns on the size of the defect relative to what n_T can resolve. Cai et al. [5, Remark 3] describe the regime in which $\hat{\theta}^{(\mathrm{meta})}$ is worse than $\hat{\theta}^{(\mathcal{T})}$, and their Assumption S4(i) bounds $\|\delta_0\|_1$ at order $\sqrt{n_T/\log d}$; Section 3.2 sweeps the defect in multiples of that radius. Where the regime begins is not fixed by $\|\delta_0\|_1$ alone, since the same norm costs an order of magnitude more concentrated on a few coordinates than spread over all of them, and none of the quantities it is stated in can be evaluated in an application. That is what the gate of Section 2.6 exists for.

The defect is unobservable, and $\hat{\delta}$ must not be read as a stand-in for it. Shrinkage makes $\|\hat{\delta}\|_1$ a lower bound, and $\hat{\delta} = (\hat{\delta}_0, 0)$ says that no correctable bias was detected at the selected penalty, not that none is present. We report $\|\hat{\delta}\|_1$ by block, its support size, and how often the slope correction is null, and read them accordingly.

Tuning. All ℓ_1 penalties are selected by stratified cross-validated log-loss. No tuning step reads the held-out target rows, but the rows a given penalty is tuned on are those its own objective is defined over: $\lambda_{\mathcal{T}}$ and λ_δ on TL alone by (6) and (9), λ_θ on SL $\cup$ TL by (8).

The last of these involves a choice about the estimand and not about numerics. The fitting weights are fixed by the objective, but nothing fixes how the validation loss should be weighted. Under the fitting weights the $n_{\mathcal{S}}$ reweighted source rows dominate the criterion, so the selected penalty minimises a source-weighted risk while the target of estimation is $E_{\mathcal{T}}$ risk. Restricting the validation loss to target rows would align the criterion with the estimand at the cost of a much noisier one when n_T is small. The weighting is an explicit switch in the implementation; the results reported here use the fitting weights throughout.

We report the selected λ_θ and λ_δ against the rates $\sqrt{n^{-1}\log d}$ and $\sqrt{n_T^{-1}\log d}$ that Cai et al. [5] assign them.

Solver, penalty grid, fold count and convergence settings are given in Supplementary Section S1.1 for the simulation and Supplementary Section S2.4 for the application.

2.5 Coalitions and the cross-fitted value function

Section 2.4 leaves two estimators of the same parameter, $\hat{\theta}^{(\mathcal{T})}$ and $\hat{\theta}^{(\mathrm{meta})}$, and because the blocks of Section 2.4 partition the slopes of θ, the choice between them can be made one block at a time.

Coalitions. For $c \subseteq \{\mathsf{X}, \mathsf{S}\}$ let $\theta^{(c)}$ take block b from $\hat{\theta}^{(\mathrm{meta})}$ when $b \in c$ and from $\hat{\theta}^{(\mathcal{T})}$ otherwise, with the intercept refitted at the resulting slopes:

$$\theta_b^{(c)} = \begin{cases} \hat{\theta}_b^{(\mathrm{meta})}, & b \in c, \\ \hat{\theta}_b^{(\mathcal{T})}, & b \notin c, \end{cases} \qquad \theta_0^{(c)} = \arg\min_{t} \sum_i \ell_{\log}\Big(y_i,\ t + x_i^\top \theta_{\mathsf{X}}^{(c)} + s_i^\top \theta_{\mathsf{S}}^{(c)} \Big), \tag{10}$$

the sum running over the rows on which the coalition is formed. We abbreviate $\{\mathsf{X}, \mathsf{S}\}$ as XS throughout. The four coalitions interpolate between the two estimators: $\theta^{(\varnothing)}$ and $\theta^{(\mathsf{XS})}$ reproduce $\hat{\theta}^{(\mathcal{T})}$ and $\hat{\theta}^{(\mathrm{meta})}$ up to that intercept, so the binary rule of Cai et al. [5, eq. 9] is recovered by comparing those two alone.

The refit is not cosmetic, and it is why the intercept sits outside both blocks to begin with. Cai et al. [5] absorb it into X; a constant column there would put a purely calibrational degree of freedom in one block and not the other, which the attribution of Section 2.7 would read as image signal. The same offset returns at the splice, since the two fits are calibrated to different populations: $\hat{\theta}^{(\mathrm{meta})}$ to a reweighted mixture of source and target, $\hat{\theta}^{(\mathcal{T})}$ to the target. Left alone, a mixed coalition would carry a prevalence difference that no block put there, and the comparison below would be partly a comparison of prevalences. Refitting removes it at the cost of one degree of freedom estimated on the same rows as the slopes.

Value function. We score coalitions by the target log loss, cross-fitted [25] over the same two-fold split of TL that Cai et al. [5, eq. 9] uses. Let F_1, F_2 be a stratified partition of $I_{\mathcal{T},L}$; the split is stratified rather than random because TL carries few positives at the sample sizes considered, and an unstratified halving can leave a fold on which the loss cannot be scored. For each k the pair $(\hat{\theta}^{(\mathcal{T})}, \hat{\theta}^{(\mathrm{meta})})$ is refitted on F_k, every coalition is assembled from that one pair by (10) with the intercept refitted on F_k, and the loss is evaluated on the complement:

$$L(c) = \frac{1}{2}\sum_{k=1}^{2} \frac{1}{|F_{3-k}|} \sum_{i \in F_{3-k}} \ell_{\log}\Big(y_i, \eta(z_i; \theta^{(c)})\Big), \qquad v(c) = L(\varnothing) - L(c). \tag{11}$$

Refitting on a fold means (6), (8) and (9) with $I_{\mathcal{T},L}$ replaced by F_k and n_T by $|F_k|$; the source rows are unchanged.

Two features of (11) matter downstream. The evaluation rows are disjoint from the rows the coalition was built on, which is what licenses reading L as a risk rather than as a fit criterion. And v is normalised so that $v(\varnothing) = 0$: every value is quoted relative to using no source data at all. The first holds for the fits of Section 2.4. On SUPERVISED the representation is itself fitted on TL (Section 4.2), so the evaluation rows reach the encoder through their own labels and L is in-sample for that branch.

The density ratio is estimated once, on unlabelled rows, and is not refitted per fold; $\hat{w}$ is applied only to source rows in (8), so the folds enter only through the labelled fits.

2.6 Transferability detection

The dead zone. A coalition is admitted only if it beats the target-only fit by a margin. One case in which ϵ_0 may be set to zero is the exact-sparse linear model with the identity link [5, Supplement S.4], where the tolerance constant τ_5 in the proof of Theorem S2 vanishes. Our link is logistic, so that case does not apply, whether or not $\bar{\theta}$ is itself sparse. For generalized linear models under the assumptions of Theorem S2, the proof uses the sufficient condition

$$\epsilon_0 \;\geq\; \left[(\kappa_{\mathrm{RE}} - \alpha_5)\big\|\hat{\theta}^{(\mathrm{meta})} - \bar{\theta}\big\|_2^2 \;+\; \tau_5 \,\frac{\log d}{n_T}\big\|\hat{\theta}^{(\mathrm{meta})} - \bar{\theta}\big\|_1^2\right]_+, \tag{12}$$

where $[a]_+ = \max(a, 0)$, $\alpha_5 > 0$ and $\tau_5 \geq 0$ are the curvature and tolerance constants of the restricted-strong-convexity (RSC) inequality, and $\kappa_{\mathrm{RE}} > 0$ is a free constant. We refer to the second summand, $\tau_5(\log d/n_T)\|\hat{\theta}^{(\mathrm{meta})} - \bar{\theta}\|_1^2$, as the RSC tolerance component of (12); (23) evaluates it at unit weight.

Nothing on the right of (12) is available in an application: $\bar{\theta}$ is unknown, so neither estimation-error norm can be computed, and the curvature constants are not identified. Nor does the condition reduce to a rate, since the explicit factor $\log d/n_T$ multiplies an unknown squared ℓ_1 error and the ℓ_2 term remains alongside it. We therefore set

$$\epsilon_0 = \frac{\log d}{n_T}, \tag{13}$$

as a schedule indexed by d and n_T, taking the explicit prefactor of the tolerance term and fixing the multiplier at one. Cai et al. [5, Supplement S.4] give no application-computable cutoff; they recommend a small positive value and note that zero may serve when the unlabelled samples are large enough. Both d and n_T are settled before any outcome is seen, so no result can move (13). Section 3.6 evaluates $\|\hat{\theta}^{(\mathrm{meta})} - \bar{\theta}\|_1$ directly, where $\bar{\theta}$ is known by construction, reports the resulting T_{RSC} of (23) as a diagnostic, and replays the decisions across a grid of thresholds.

A threshold chosen so that the null acceptance rate falls below a nominal level would make that rate an artefact of the choice rather than an observation. We compute and report the corresponding null-derived cutoffs as diagnostics, but the threshold driving the decisions comes from (13).

Two rules. Cai et al. [5, eq. 9] compare $\hat{\theta}^{(\mathcal{T})}$ with $\hat{\theta}^{(\mathrm{meta})}$ and take the source whole or not at all. Four coalitions let the same comparison retain the source for one block alone, since a single-block coalition can remain admissible when the joint one is not. The two rules are

$$\hat{\rho} = \mathbf{1}\big\{L(\mathsf{XS}) \leq L(\varnothing) - \epsilon_0\big\}, \qquad \hat{c} = \arg\min_{c \in \mathcal{E}} L(c), \quad \mathcal{E} = \{\varnothing\} \cup \big\{c : L(c) \leq L(\varnothing) - \epsilon_0\big\}. \tag{14}$$

The dead zone requires a coalition to improve on the target-only fit by a non-zero margin before displacing it, and does not separate admitted coalitions from one another. In the original proof that margin offsets the restricted-strong-convexity tolerance; here it is not calibrated to the sampling distribution of the fold-loss differences. Both rules read the same threshold and the same four losses, though $\hat{\rho}$ reads only two of them. The deployed model is $\hat{\beta}(\theta^{(\hat{c})})$, with $\hat{\rho}$ retained as a comparator.

What carries over. On the criterion the two rules share, the coalition rule cannot do worse than the candidate the binary rule selects. Write $\hat{c}_{\mathrm{bin}} = \mathsf{XS}$ when $\hat{\rho} = 1$ and $\hat{c}_{\mathrm{bin}} = \varnothing$ otherwise. Since $\mathcal{E}$ always contains $\varnothing$, and contains XS whenever $\hat{\rho} = 1$, we have $\hat{c}_{\mathrm{bin}} \in \mathcal{E}$ and therefore

$$L(\hat{c}) \;\le\; L(\hat{c}_{\mathrm{bin}}) \tag{15}$$

identically, with no assumptions. This is dominance on the selection criterion, not on held-out performance.

Equation (15) should not be conflated with the estimation-error guarantee of Cai et al. [5, Theorem 2]. Extending that theorem to the coalition rule would require uniform control of every spliced candidate under analogues of the original sparsity, regularity and dead-zone conditions. Cross-fitting makes the data a candidate is fitted on independent of the fold its loss is evaluated on, and admission supplies the required margin over the target-only loss. For the slopes, writing $\theta_{-0} = (\theta_{\mathsf{X}}^\top, \theta_{\mathsf{S}}^\top)^\top$,

$$\big\|\theta_{-0}^{(c)} - \bar{\theta}_{-0}\big\|_1 \;\le\; \big\|\hat{\theta}_{-0}^{(\mathcal{T})} - \bar{\theta}_{-0}\big\|_1 + \big\|\hat{\theta}_{-0}^{(\mathrm{meta})} - \bar{\theta}_{-0}\big\|_1, \tag{16}$$

whose two right-hand terms are controlled, under the corresponding assumptions, by the original meta-learning result and the standard target-only rate. The intercept refitted on F_k by (10) and the data-dependent selection over the candidate set (four elements here, 2^B with B blocks) are not covered. We do not establish the extension, so (15) is what this section claims and it holds without assumptions.

2.7 Block attribution

Resolving the source contribution by block yields two things that are easily conflated. One is a finer *selection* rule: the procedure may retain the source-borrowing fit for one block while keeping the target-only fit for the other, where the binary rule can only accept or refuse both. That needs no more than an ordering of the coalition losses. The other is an *attribution*: a pair of scalars ϕ_{X} and ϕ_{S} summarising how much each block contributed to whatever gain the source delivered. This asks more, since the two scalars must be comparable across blocks, across representation branches and across repeated splits, and must sum to the total gain. The definition below gives them the summation property; what they do and do not measure is stated after it.

Definition. Treating the two blocks as players and v of (11) as the characteristic function of a cooperative game, the Shapley values [9] are

$$\phi_{\mathsf{X}} = \tfrac{1}{2}\big[\{v(\mathsf{X}) - v(\varnothing)\} + \{v(\mathsf{XS}) - v(\mathsf{S})\}\big], \qquad \phi_{\mathsf{S}} = \tfrac{1}{2}\big[\{v(\mathsf{S}) - v(\varnothing)\} + \{v(\mathsf{XS}) - v(\mathsf{X})\}\big], \tag{17}$$

each block's marginal contribution averaged over the two orders in which the blocks may be added. With two players the average is over $2! = 2$ permutations, so (17) is exact: no sampling is involved, and the approximation error that attends Shapley values in high-dimensional feature attribution is absent here. Note that (17) reads only the four coalition losses and never ϵ_0; the attribution is unaffected by the threshold and by which coalition is deployed.

Proposition 1 (Efficiency)**.** $\phi_{\mathsf{X}} + \phi_{\mathsf{S}} = v(\mathsf{XS}) = L(\varnothing) - L(\mathsf{XS})$.

This follows from (17) and the normalisation $v(\varnothing) = 0$ by inspection. The total gain is fully allocated between the two blocks, with the synergy $v(\mathsf{XS}) - v(\mathsf{X}) - v(\mathsf{S})$ split evenly and nothing left over. A reported residual is a numerical consistency check on the implementation, not evidence about the decomposition. The intercept is not a player: it is not one of the components c ranges over, and every coalition refits it unconditionally by (10), so there is no borrowing decision to attribute. Its effect enters each $L(c)$, and hence each ϕ_b, but not as a share of them.

What ϕ_b measures. ϕ_b is the marginal value of borrowing block b from the source, evaluated on target risk. Three things follow.

1. It carries a bias term and a variance term with opposite signs, not separately identified from the four coalition losses, so only their sum is signed by the data and only the first is transferability.
2. It is defined on θ, and by (1) a coalition reaches $\hat{\beta}$ only through the imputed labels of (19).
3. Nothing in the construction requires the ordering to hold replicate by replicate at a fixed design.

Section 3 takes up all three on data where δ_0 is known by construction; the block-strength diagnostic that the first calls for is reported alongside the attribution in Section 4.3.

Proposition 2 (The attribution gap is a coalition-loss difference)**.** $\phi_{\mathsf{S}} - \phi_{\mathsf{X}} = L(\mathsf{X}) - L(\mathsf{S})$.

This constrains how the third of the three can be tested. The two-block and target-only losses cancel from the gap, so a log-loss contrast between those same two coalitions is a second estimate of the functional Proposition 2 is built from rather than an independent check on it. Any test of whether the attribution tracks realised performance must use a different metric; Section 3.7 uses AUC and the Brier score [27], the latter being the proper scoring rule closest to the loss v is built on.

Two layers. Finally, (11) is a loss on θ, which reads all of Z, whereas the endpoints of Table 1 are computed from the estimator $\hat{\beta}$ of Section 2.8, which reads X alone. Writing $\varpi = n_T/(n_T + N_T)$ for the labelled fraction of the target sample and $s_\beta = \|\beta_0\|_0$, Cai et al. [5, Theorem 3] connects the two,

$$\|\hat{\beta} - \beta_0\|_2^2 = O_p\left((1-\varpi)\big\|\hat{\theta} - \bar{\theta}\big\|_2^2 + s_\beta \frac{\log p}{n_T + N_T}\right), \tag{18}$$

but through the estimation error of θ, not through its discrimination. Two coalitions may impute equally well and yield very different $\hat{\beta}$ if their errors sit in different blocks, since only the part of $\hat{\theta} - \bar{\theta}$ lying in θ_{X} has a counterpart in β. The attribution and the endpoint are therefore not guaranteed to agree, and nothing in (17) makes them agree. Section 3 takes up the question directly.

2.8 Outcome model

Given an imputation parameter $\hat{\theta}$, the target unlabelled rows receive soft labels $\hat{m}(z_i) = g\{\eta(z_i; \hat{\theta})\}$, the labelled target rows keep their observed outcomes, and the outcome model is fitted on images alone:

$$(\hat{\alpha}, \hat{\beta})(\hat{\theta}) = \arg\min_{\alpha,\beta} \Big\{ -\frac{1}{n_T + N_T} \Big[\sum_{i \in I_{\mathcal{T},U}} Q\big\{g(\alpha + x_i^\top \beta), \hat{m}(z_i)\big\} + \sum_{i \in I_{\mathcal{T},L}} Q\big\{g(\alpha + x_i^\top \beta), y_i\big\} \Big] + \lambda_\beta \|\beta\|_1 \Big\}, \tag{19}$$

the empirical counterpart of (1). The labelled rows are a fraction ϖ of the total, which is small in our setting, so the fit is dominated by imputed rows: at this stage the accuracy of $\hat{\theta}$ matters more than n_T does, and (18) is the formal version of that statement. As in Section 2.4, λ_β is selected by cross-validation on the rows (19) is fitted over, so on TU the validation target is $\hat{m}(z_i)$ rather than an observed outcome. No binarisation of the soft label is required, and none is applied; Cai et al. [5, Section S.10] report that thresholding at the target prevalence performed slightly worse.

Two features of (19) bear on the rest of the paper. First, the soft labels inherit the calibration of $\hat{\theta}$ through its intercept, so the refit in (10) propagates here: a coalition whose intercept was left as inherited would impute at the wrong prevalence, and the resulting shift would be absorbed by $\hat{\alpha}$ rather than showing up as a discrimination loss. Second, no part of (19) enters the coalition losses of Section 2.5. Selection happens entirely upstream of the outcome model, on a criterion evaluated at θ, while the endpoints are evaluated here; that the two need not agree is the point taken up in Section 3.

The deployed estimator is $\hat{\beta}(\theta^{(\hat{c})})$, but we fit (19) once for each of the four coalitions, so that every reported estimator is complete rather than a re-scoring of a shared fit. This yields the six endpoints of Table 1.

Table 1: The six endpoints. Each is $\hat{\beta}(\theta^{(c)})$ from (19) for the coalition in the second column; the coalitions are those of (10) and the two selection rules are those of (14).

Endpoint	Coalition c
TARGET-ONLY	$\varnothing$
BORROW-X	X
BORROW-S	S
CORRECTED-META	XS
GATE-BINARY	XS if $\hat{\rho} = 1$, else $\varnothing$
GATE-BLOCK	$\hat{c}$

GATE-BINARY is the endpoint of $\hat{\rho}$: the decision of Cai et al. [5, eq. 9] as it runs inside the present pipeline, and therefore a within-pipeline baseline rather than a reproduction of their reported estimator. Items 3 and 4 of Section 2.2, together with the weight clipping of Section 2.3 and the intercept refit of Section 2.5, give the respects in which that implementation departs from theirs. GATE-BLOCK is the endpoint of $\hat{c}$ and is what this paper adds.

The names refer to the fitted outcome model; where the imputation parameter itself is the object, as in (11) and in Section 3, we write $\theta^{(c)}$. External comparators, which are not part of the construction, are defined in Section 4.2.

2.9 Evaluation principles

Three choices govern every number reported in Sections 3 and 4. They are stated here because they apply to both.

Three metrics. We report the area under the ROC curve, the Brier score [27] and the expected calibration error [26], and read them together. A coalition splices blocks from two fits calibrated to different populations, so its characteristic failure is a calibration offset rather than a loss of discrimination, and by (19) such an offset propagates to the imputed labels and is then absorbed by $\hat{\alpha}$. AUC is blind to it and the Brier score confounds it with discrimination, so the calibration error is what makes the failure mode of the splice visible. The refit in (10) and the three-metric report are parts of one decision.

Evaluation is out of sample. By (19) the outcome model is fitted on $I_{\mathcal{T},U}$ together with $I_{\mathcal{T},L}$, so neither set is available for evaluation: the labels on $I_{\mathcal{T},U}$ are imputed rather than observed, but the optimisation is carried out on those rows all the same, and $\hat{\theta}$ has already read $I_{\mathcal{T},L}$. Performance is therefore computed on the held-out labelled target rows of Section 2.1, which enter no fitting or tuning step.

Contrasts, not levels. At the sample sizes considered, the level of any single held-out figure is governed largely by which rows fell into the held-out set. The six endpoints of Table 1 share $\hat{w}$, the fold construction and the single pair $(\hat{\theta}^{(\mathcal{T})}, \hat{\theta}^{(\mathrm{meta})})$ within each fold, and differ only in which c reaches (19); a difference between two of them, computed on the same split, is therefore attributable to the coalition, while the level of either is not. Every claim about the construction is accordingly a paired contrast against TARGET-ONLY on a common split, aggregated over repeated splits, and the levels are reported only to situate them.

The external comparators of Section 4.2 share none of those stages, so a contrast against them is evidence that the construction helps or does not, never an attribution to a block; CEILING and CNN are reported as levels only. Repetitions, aggregation and uncertainty are given in Sections 3.1 and 4.1.

3 Simulation studies

3.1 Design

The simulation has two halves. Sections 3.2 to 3.4 vary one thing each: where the transferability defect sits and how sparse it is, how strong the two blocks are relative to one another, and how much labelled target data there is. Sections 3.5 to 3.7 add no design axis and put three questions to the replicates those sweeps produced: what the coalition rule buys over the binary one, whether ϵ_0 is placed where it should be, and what ϕ measures.

Generating process. We follow Cai et al. [5, Sec. 4] and add block structure to the shift. A latent Gaussian matrix with AR(1) correlation $\nu = 0.1$ is passed through $t \mapsto \lfloor \log(1 + e^t) \rfloor$ to give the image block; the first three surrogate columns are built from overlapping windows of the first ten image columns, coupling the two blocks, and the rest are transformed latent draws. After standardisation Z has $p = 128$ image and $q = 93$ surrogate coordinates (79 term features, 14 metadata), so $d = 221$. R is drawn from a logistic model in Z rather than assigned, so Section 2.3 faces the kind of overlap the application does. The reference cell uses $n_T = 200$, $n_{\mathcal{S}} = 1200$, $N_T = N_{\mathcal{S}} = 10^4$ and 200 held-out labelled target rows.

Writing $\bar{\theta}$ for the target population maximiser of Section 2.4 and $\bar{\theta}^{\mathcal{S}}$ for its counterpart under $\mathcal{S}$, outcomes are drawn as $Y \mid Z \sim \mathrm{Bernoulli}[g\{\eta(Z; \bar{\theta})\}]$ on target rows and with $\bar{\theta}^{\mathcal{S}}$ on source rows, so

$$\Delta = \bar{\theta}^{\mathcal{S}} - \bar{\theta} \tag{20}$$

is a failure of (2) we construct. The working model is correctly specified throughout. The intercept is set by bisection to put the target event rate at the application's 0.105; Δ carries no intercept component and the source rate is left free.

Scenarios. Each scenario fixes the support of Δ:

$\mathcal{D}_{\varnothing}$	$\Delta = 0$; (2) holds
$\mathcal{D}_{\mathsf{X}}$	Δ on θ_{X}: the image block does not transfer
$\mathcal{D}_{\mathsf{S}}$	Δ on θ_{S}: the surrogate block does not transfer
$\mathcal{D}_{\mathsf{XS}}$	Δ on both blocks, swept in magnitude
$\mathcal{D}_{\mathrm{dense}}$	Δ spread over every coordinate of both blocks

$\mathcal{D}_{\mathsf{X}}$ and $\mathcal{D}_{\mathsf{S}}$ place Δ on seven coordinates of the named block, $\mathcal{D}_{\mathsf{XS}}$ on four of each, $\mathcal{D}_{\mathrm{dense}}$ on all 118 free image and 90 free surrogate coordinates with weights decaying as $1/\sqrt{\mathrm{rank}}$. The ten image and three surrogate columns carrying the coupling are excluded, so a shift on one block cannot reach the other through the covariates. $\mathcal{D}_{\mathrm{dense}}$ is the regime the application sits in, its features being learned representations in which signal is distributed rather than sparse; Cai et al. [5, Remark 3] gives the condition for negative transfer there in terms of rates, and this scenario is its finite-sample counterpart.

Ground truth and its calibration. What the method responds to is $\delta_0 = \bar{\theta} - \theta^*$ of (7), not Δ, and we compute it rather than approximate it. Under a correct density ratio the tilted source term in (7) is an expectation over the target covariate distribution carrying the source conditional mean, so θ^* is the projection onto the working model of $a\, g\{\eta(Z;\bar{\theta}^{\mathcal{S}})\} + (1-a)\, g\{\eta(Z;\bar{\theta})\}$ under $Z \sim \mathcal{T}$, and $\bar{\theta}$ the projection of $g\{\eta(Z;\bar{\theta})\}$. Both are unpenalised logistic fits against a soft label on 5000 unlabelled target rows.

Δ is scaled in units of $\sqrt{n_T/\log d}$, the rate at which Assumption S4(i) of Cai et al. [5] bounds the defect's ℓ_1 radius; at the reference cell it is 6.09. The linearisation $\delta_0 \approx -a\Delta$ does not deliver that norm, and its error is not a constant one could divide out: it peaks near one radius and falls as the logit saturates, and is exact only under $\mathcal{D}_{\mathrm{dense}}$ (Supplementary Table S3). We therefore solve for the scale at which the realised $\|\delta_0\|_1$ hits the requested multiple. That also gives $\mathcal{D}_{\mathsf{X}}$ and $\mathcal{D}_{\mathsf{S}}$ identical $\|\delta_0\|_1$ in every replicate, so a row of the ladder in Section 3.3 changes only which block holds the defect.

A mixture of links is not the link of a mixture, so some of δ_0 always leaks into the block the scenario left alone; each replicate is labelled from its own realised δ_0 rather than from its scenario.

What is estimated and where it is scored. The estimator is Algorithm 1, reimplemented so that the simulation runs the same logic as the application pipeline without importing it, and scored under Section 2.9. ϵ_0 is recomputed per cell, since the grid sweeps n_T. Both layers of Section 2.7 are scored in every replicate: $\theta^{(c)}$, which is what ϕ is defined on, and $\hat{\beta}(\theta^{(c)})$, fitted separately for each coalition rather than re-scored from a shared fit. So are both rules of (14), but they are reported differently. The GATE column of Sections 3.2 to 3.4 is GATE-BLOCK, while $\hat{\rho}$ appears there as an acceptance rate rather than as a fitted endpoint: those sections ask where transfer stops paying and where the binary rule stops accepting it. It is a share of the replicates at a design point, since (14) reads the cross-fitted losses of the realised sample, so a single δ_0 yields both acceptances and refusals. The two rules are compared as estimators in Section 3.5, and again in the application where δ_0 is unknown (Section 4.3).

Replication and aggregation. Every scenario–magnitude point carries 50 replicates. Contrasts are summarised by their mean with a paired t-interval, decisions and identifications by their proportion, and the agreement between ϕ and realised performance by a Spearman correlation across the replicates of a cell at fixed magnitude, since pooling magnitudes would let between-cell variation in δ_0 stand in for the replicate-level agreement at issue.

What was run. Table 2 lists the runs, with block-strength levels as defined in Section 3.3. The two ablations are paired re-runs of the reference cell on the same covariates, splits and outcome variates: the oracle imputation of Section 3.7, and a held-out correction that fits (9) on target rows the meta fit did not see. The second enters the checks of Supplementary Section S1.3 and is not reported as a result of its own. All use correctly specified nuisance models; the misspecified variants are in the implementation but not reported here.

Table 2: Simulation runs; magnitudes are multiples of $\sqrt{n_T/\log d}$, and block-strength levels are as defined in Section 3.3.

Run	Scenarios	Magnitudes	Reps	Read in
negative-transfer axis	$\mathcal{D}_{\varnothing}, \mathcal{D}_{\mathsf{XS}}$	0; 0.5–3 (7 values)	400	§3.2, §3.6
dense defect	$\mathcal{D}_{\mathrm{dense}}$	0.5, 1, 1.5, 2	200	§3.2
held-out correction	$\mathcal{D}_{\mathsf{XS}}, \mathcal{D}_{\mathrm{dense}}$	0.5, 1, 1.5, 2	400	§S1.3
block strength, 6 levels	$\mathcal{D}_{\mathsf{X}}, \mathcal{D}_{\mathsf{S}}$	0.5, 1, 1.5, 2	2400	§3.3, §3.5
n_T sweep, $\{100, \ldots, 800\}$	$\mathcal{D}_{\mathsf{X}}, \mathcal{D}_{\mathsf{S}}$	1	600	§3.4, §3.5, §3.6
oracle imputation	$\mathcal{D}_{\varnothing}, \mathcal{D}_{\mathsf{XS}}$	0; 0.5–2 (6 values)	350	§3.7
Total			4350	

When n_T is swept, $\|\delta_0\|_1$ is anchored at the reference cell's radius: the application's δ_0 is whatever it is, and the question is whether more labelled target data resolves it.

Design noise. The target prevalence is held at 0.105, but Δ carries no intercept component, so the source rate is free to move as the source slopes shift. It does, and furthest on the negative-transfer axis, where the largest magnitude pairs a target rate of 0.105 with a source rate of 0.313: the far end of that axis carries a shift in prevalence as well as one in slopes. Supplementary Section S1.2 gives the source rate and the realised positive counts for every cell.

Numerical checks. Three properties hold across all $4{,}350$ replicates and are not revisited below: Proposition 1 to machine precision; no replicate whose dominance label is changed by the leakage above; and the calibration of $\|\delta_0\|_1$ reaching the requested norm wherever the link can express it. Supplementary Section S1.3 gives the figures. A fourth, the paired re-solve of Supplementary Section S1.3.4, leaves $\hat{\rho}$ and $\hat{c}$ unchanged on the 172 pairs of Supplementary Table S4. Equation (15) holds on every replicate of $\mathcal{D}_{\mathsf{X}}$ and $\mathcal{D}_{\mathsf{S}}$.

Where the simulation sits. The reference cell matches the application on n_S, target prevalence, the difficulty of the density-ratio problem, and on n_T, p and q in its main arm and its two DenseNet branches. It does not match it on the absolute discrimination either block affords.

3.2 The negative-transfer axis and its calibration

This section asks where transfer stops paying and where the gate stops accepting it, and whether the two coincide. Neither question can be put to real data: Section 2.4 states the regime in which borrowing stops paying in quantities no application can evaluate, and the gate of Section 2.6 stands in for them there. We sweep $\mathcal{D}_{\mathsf{XS}}$ from half a radius $\sqrt{n_T/\log d}$ to three at the reference cell. The axis means what it says: the realised $\|\delta_0\|_1$ lies within 0.6% of the requested multiple at every magnitude on this axis, and under $\mathcal{D}_{\varnothing}$ the computed δ_0 is zero to fifteen decimal places, so the projections that define it introduce no defect of their own (Supplementary Section S1.3.3).

Table 3: The axis at both the selection and the deployed layer, over the 50 replicates at each magnitude. $\hat{c}$ counts the coalition of (14). $L(\varnothing) = 0.334$ at every magnitude, so the margin carries the whole of the criterion. The requested three radii realise as 2.98 (Supplementary Table S3).

	$\|\delta_0\|_1$	$\hat{c}$				$\hat{\rho}$	$L(\mathsf{XS})$	AUC		
Scenario	(radii)	$\varnothing$	X	S	XS	rate	$-\,L(\varnothing)$	T-only	C-meta	Gate
$\mathcal{D}_{\varnothing}$	0	2	0	0	48	0.96	$-.115$	.650	.799	.794
$\mathcal{D}_{\mathsf{XS}}$	0.50	5	1	0	44	0.88	$-.099$	.650	.797	.783
	0.75	8	3	1	38	0.82	$-.087$	.650	.783	.751
	1.00	11	2	0	37	0.76	$-.066$	.650	.768	.741
	1.25	17	2	4	27	0.64	$-.045$	.650	.756	.701
	1.50	29	1	5	15	0.34	$-.006$	.650	.741	.671
	2.00	40	4	3	3	0.10	$+.049$	.650	.708	.661
	3.00	49	0	0	1	0.02	$+.166$	.650	.644	.654
$\mathcal{D}_{\text{dense}}$	0.50	1	0	1	48	0.96	$-.112$	.650	.799	.789
	1.00	1	0	1	48	0.98	$-.109$	.650	.797	.787
	1.50	2	0	1	47	0.94	$-.108$	.650	.794	.783
	2.00	2	3	0	45	0.94	$-.102$	.650	.790	.778

The gate tightens monotonically, above a floor of false refusals. Fallbacks to $\varnothing$ rise without a reversal across the axis and $\hat{\rho}$ falls with them (Table 3). What the rate leaves out is its own floor: under $\mathcal{D}_{\varnothing}$, where δ_0 is zero to fifteen decimal places and the margin runs four times ϵ_0, the gate still refuses twice in fifty. That is the false-refusal rate at $n_T = 200$, and it is the cost of deciding on the realised sample.

The radius measures the estimation problem, not the damage. Spread across both blocks rather than concentrated in a handful of coordinates, the same $\|\delta_0\|_1$ barely touches the estimator. The gate goes on accepting almost every replicate out to two radii, $\hat{c} = \mathsf{XS}$ throughout, and the margin holds where $\mathcal{D}_{\mathsf{XS}}$ has already crossed into positive. On the deployed model CORRECTED-META gives up .009 AUC across the whole axis, against .089 over the same range under $\mathcal{D}_{\mathsf{XS}}$.

Two things happen at once when the defect spreads, and they pull the same way. The correction of (9) loses its grip: recovering a defect spread over k coordinates needs

$$\|\delta_0\|_1 \;>\; k\lambda_\delta, \qquad k\lambda_\delta \Big/ \sqrt{n_T/\log d} \;=\; \begin{cases} 0.22, & k = 8, \\ 5.97, & k = d = 221, \end{cases} \tag{21}$$

so the dense scenario asks for a defect larger by a factor of $d/8$ before the correction engages. Both figures use λ_δ at its assigned order $\sqrt{\log d/n_T}$, and cross-validation selects it below that rate (Supplementary Table S13); the same λ_δ enters both, so the ratio is unaffected. Here $k = d$ bounds the 208 free coordinates the dense design fills. But the defect also loses its bite. What it costs is $\delta_0^\top \Sigma \delta_0$, and at fixed $\|\delta_0\|_1$ that quadratic form shrinks as the defect spreads,

$$\|\delta_0\|_2^2 \asymp \|\delta_0\|_1^2/k, \tag{22}$$

by an order of magnitude between the two scenarios. The second effect is the larger, which is why the estimator survives a defect the correction cannot touch.

So the radius orders difficulty within a sparsity level but not across them. The correction reaches a sparse defect and not a dense one, while the damage runs the other way, so $\mathcal{D}_{\mathsf{XS}}$ is the adverse case here and $\mathcal{D}_{\text{dense}}$ the mild one, and the crossings in Table 4 are the earliest this axis produces.

Table 4: Where CORRECTED-META falls behind TARGET-ONLY, in radii. At the deployed layer it is behind on the calibration error at every magnitude including $\mathcal{D}_\varnothing$, so there is no crossing to report.

	Imputation layer	Deployed layer
ECE	0.15	behind throughout
Brier	1.72	1.94
AUC	2.89	2.89

Negative transfer distorts scale before ordering. δ_0 distorts the scale of the coefficients more than the ordering they induce.

Across metrics. The three metrics cross in the order they read that scale, calibration first and AUC last (Table 4), and under $\mathcal{D}_{\text{dense}}$ neither the Brier score nor the AUC crosses anywhere on the axis (Supplementary Table S6). Equation (11) is a log loss and sits at the sensitive end of that range, so the gate reads the earliest of the three warnings, and the conservatism above follows from the criterion. Where the downstream use is ranking, the gate will decline transfers that would still have served: $\hat{\rho}$ passes 0.5 at 1.37 radii, and at 1.50 radii it gives up 0.070 AUC against CORRECTED-META.

Across layers. The part of δ_0 sitting in θ_{S} reaches $\hat{\beta}$ only through the imputed labels of (19), since by (1) the deployed model reads X alone. What is attenuated on the way is the scale distortion, not the ordering: the Brier crossing comes earlier at the imputation layer while the AUC crossing coincides (Table 4). $\mathcal{D}_{\text{dense}}$ supplies the other half of that argument. There the defect distorts the scale hardly at all, the deployed calibration error of CORRECTED-META is flat along the axis where $\mathcal{D}_{\mathsf{XS}}$ raises it by about a quarter, and the two layers separate nowhere (Supplementary Table S6). Section 3.7 takes up the same restriction on the surrogate block's benefit.

3.3 The block-strength ladder

The ladder has six rungs, indexed by the gap in held-out AUC between a model using the surrogate block alone and one using the image block alone, running from image-dominant to surrogate-dominant; the four middle rungs span the range the application's three branches occupy (Supplementary Section S2.7). Every rung carries $\mathcal{D}_{\mathsf{X}}$ and $\mathcal{D}_{\mathsf{S}}$ at the same four magnitudes, so across a row the only thing that changes is which block holds the defect.

Table 5: Identification is the share of the 200 replicates in each cell in which ϕ is smaller for the block carrying δ_0. Sums are computed from the unrounded rates; the *surrogate, soft* row is the one where this shows.

	AUC gap	Identification		
Rung	$\mathsf{S}-\mathsf{X}$	$\mathcal{D}_{\mathsf{X}}$	$\mathcal{D}_{\mathsf{S}}$	Sum
image-dominant	-0.344	0.09	1.00	1.09
balanced	-0.004	0.82	0.82	1.64
surrogate, soft	$+0.075$	0.93	0.65	1.57
surrogate, medium	$+0.124$	0.99	0.49	1.48
surrogate, strong	$+0.186$	0.99	0.37	1.36
surrogate-dominant	$+0.331$	1.00	0.10	1.10

Identification responds to two things at once. Across Table 5 the two rates run in opposite directions and cross where the blocks are equally strong: a defect in the image block is almost never found when that block is the stronger and always found when it is the weaker, and the sum falls from its maximum at the balanced rung to near one at either end.

Table 6: The three surrogate rungs of Table 5, *soft* to *strong*, under $\mathcal{D}_{\mathsf{S}}$, with gap $= \phi_{\mathsf{S}} - \phi_{\mathsf{X}}$; the last two columns are linear interpolations between adjacent magnitudes.

		$\|\delta_0\|_1$ (radii)				Gap reaches	Ident. reaches
Rung		0.5	1.0	1.5	2.0	zero at	0.5 at
surrogate, soft	gap	$+.021$	$-.016$	$-.046$	$-.094$	0.78	0.83
	ident.	.30	.60	.74	.94		
surrogate, medium	gap	$+.042$	$+.014$	$-.026$	$-.086$	1.17	1.29
	ident.	.16	.34	.62	.82		
surrogate, strong	gap	$+.072$	$+.047$	$-.000$	$-.047$	1.50	1.57
	ident.	.10	.16	.46	.74		

Within a rung, Table 6 shows the gap $\phi_{\mathsf{S}} - \phi_{\mathsf{X}}$ starting positive and turning negative as $\|\delta_0\|_1$ grows, with identification crossing 0.5 a little after the gap crosses zero.

The gap $\phi_{\mathsf{S}} - \phi_{\mathsf{X}}$ is a sum of two terms, one set by the strength imbalance and unrelated to δ_0, the other carrying the position and size of δ_0. The identification criterion reads only the sign of the sum, so the two cannot be separated. What the ladder measures is the first: δ_0 has to cancel the imbalance before it shows, and across the three rungs the radius at which it does rises with the strength gap at about six radii per unit of AUC. The ordering of ϕ across blocks is therefore interpretable only where the blocks are comparably strong.

3.4 Sample size: identification and tracking under a sweep of n_T

Sections 3.2 and 3.3 vary the transferability defect while holding n_T fixed. Here we instead hold its absolute magnitude fixed at $\|\delta_0\|_1 = 6.09$ and vary the labelled target sample size, which is what separates the two readings a null tracking result admits: finite-sample noise, or a property of the decomposition. Expressed in units of the reference radius $\sqrt{n_T / \log d}$, the fixed defect decreases from 1.41 radii at $n_T = 100$ to 0.50 at 800. Under the prespecified operational schedule (13), the gate threshold is recomputed at each sample size and shrinks eightfold over the same range.

Table 7: $\hat{\rho}$ is pooled over the two scenarios; the single-block columns count, out of 100, replicates selecting the uncontaminated and the contaminated block. $|\rho_s|$ is the median over the eight metric-by-scenario Spearman correlations at a cell.

	$\|\delta_0\|_1$		Identification		Gap			Single block chosen		
n_T	abs.	radii	$\mathcal{D}_{\mathsf{X}}$	$\mathcal{D}_{\mathsf{S}}$	$\mathcal{D}_{\mathsf{X}}$	$\mathcal{D}_{\mathsf{S}}$	$\hat{\rho}$	clean	contam.	$\|\rho_s\|$
100	6.09	1.41	.66	.72	$+.071$	$-.056$	0.55	23	1	.084
200	6.09	1.00	.76	.80	$+.047$	$-.054$	0.79	26	1	.091
300	6.09	0.82	.88	.76	$+.050$	$-.026$	0.95	21	0	.047
400	6.09	0.71	.84	.80	$+.040$	$-.032$	0.91	18	0	.117
500	6.09	0.63	.86	.90	$+.030$	$-.030$	0.91	20	0	.099
800	6.09	0.50	.86	.92	$+.022$	$-.027$	0.73	**48**	2	.063

n_T separates two quantities (Table 7). Identification improves across the sweep, and does so against the axis: the defect shrinks in radii as n_T grows, and by Section 3.3 the smaller multiple is the harder one. The replicate-level agreement between ϕ and realised performance shows no such trend; $|\rho_s|$ pools to $.0745$ over the 48 correlations, and 3 of them reach $p < 0.05$ against the 2.4 expected by chance. Section 3.7 resolves the same correlations by layer and metric over the full set of cells.

The selection rule names the contaminated block more reliably than the ordering of ϕ does, and less often. It commits to a single block on about a fifth of the replicates up to $n_T = 500$ and on half of them at 800, while the ordering commits every time. What n_T buys the selection rule is coverage rather than accuracy: its precision is at least 96% at every cell and does not trend, and what doubles at $n_T = 800$ is the share of replicates on which it answers at all.

3.5 The two rules compared

Sections 3.2 to 3.4 read $\hat{\rho}$ as an acceptance rate and $\hat{c}$ as a selection, but never against each other. Here they are compared on the same replicate, on the scenarios built for the comparison and at a δ_0 that is known: the $3{,}000$ replicates of $\mathcal{D}_{\mathsf{X}}$

and $\mathcal{D}_{\mathsf{S}}$ carried by the ladder of Section 3.3 and the sweep of Section 3.4. No design axis is added; the question is put to replicates already reported.

The sweep holds $\|\delta_0\|_1$ fixed in absolute terms, so its n_T axis runs down the radius axis rather than across a second one, and both batches are read on the radius axis below. Resolved by n_T instead, the comparison is flat at zero: coverage doubles by $n_T = 800$, but the defect is smallest there and both departures shrink to nothing with it.

$\hat{c}_{\mathrm{bin}}$ takes values in $\{\varnothing, \mathsf{XS}\}$ and $\hat{c}$ in all four coalitions, so the two rules can part only where $\hat{c}$ names a single block, and two departures exhaust the disagreement: a narrowing, with $\hat{\rho} = 1$ and $\hat{c} \in \{\mathsf{X}, \mathsf{S}\}$, and a rescue, with $\hat{\rho} = 0$ and the same $\hat{c}$. The rules agree on 58.3% of the replicates, narrow on 24.6% and rescue on 17.1%. Both departures pay on the criterion the two rules share, by (15): a narrowing improves the cross-fitted loss by .026 on average and a rescue by .064. Of the two blocks a departure rewrites, (1) reads only the image block at deployment, and which way that block can move is fixed by $\hat{c}_{\mathrm{bin}}$: a rescue can supply θ_{X} and never withdraw it, a narrowing can withdraw it and never supply it.

What the deployed difference tracks. Table 8 splits the departures by what the coalition rule does to each block relative to $\hat{c}_{\mathrm{bin}}$. The deployed column follows the image block: large on the two rows where θ_{X} moves, signed by whether it is supplied or withdrawn, and near zero on the two where it does not, the narrowing row over an interval covering zero. It follows θ_{S} nowhere, because by (1) the deployed model reads X alone: a commitment about θ_{S} reaches $\hat{\beta}$ only through the imputed labels. The imputation column, where θ_{S} is scored on its own terms, splits by departure instead, a rescue being measured against a binary rule holding neither block and a narrowing against one holding both. Which block a departure names therefore separates the two at neither layer: narrowings name S on .61 of the occasions they occur and rescues on .69. What separates them is what the commitment does to the block the estimand reads.

Table 8: The two departures split by what the coalition rule does to each block relative to $\hat{c}_{\mathrm{bin}}$: supplied where it takes the meta value the binary rule withholds, withdrawn where it returns to the target-only value the binary rule transfers, unchanged where the two rules agree. Differences are GATE-BLOCK minus GATE-BINARY on the same replicate; positive favours the coalition rule.

			Effect on		ΔAUC	
Departure	$\hat{c}$	n	θ_{X}	θ_{S}	deployed	imputation
narrowing	X	289	unchanged	withdrawn	$+.000$	$+.001$
rescue	S	356	unchanged	supplied	$+.007$	$+.136$
rescue	X	158	supplied	unchanged	$+.086$	$+.075$
narrowing	S	449	withdrawn	unchanged	$-.042$	$+.002$

Pooled over the whole range the deployed difference is $-.001$; resolved by radius in Table 9 it rises, and changes sign near the interval in which the gate begins to refuse (Section 3.2). Which of the two departures is on offer is not the coalition rule's choice. While $\hat{\rho}$ still accepts, the only move open to it is to narrow; once $\hat{\rho}$ begins to refuse, the only move open is to rescue. The acceptance rate falls across the range, so the composition of the departures shifts with it, and with the composition the direction in which the image block moves. The value of either departure does not change with the radius: the gain on a rescue is flat, with a rank correlation of $-.04$ ($p = .37$). What rises is how often the paying move is the one on offer. A rescue is the coalition rule keeping source information on a replicate where the binary rule deploys $\hat{\beta}(\theta^{(\varnothing)})$ and keeps none, and at two radii it does so on more than four in ten of them.

Table 9: The same differences by the realised $\|\delta_0\|_1$ in units of $\sqrt{n_T/\log d}$. On the replicates where the rules agree the difference is zero by construction, so the two departure columns carry the whole of the pooled figure.

			Narrowing		Rescue		Pooled ΔAUC	
$\|\delta_0\|_1$ (radii)	N	$\hat{\rho}$ rate	n	ΔAUC	n	ΔAUC	deployed	imputation
0.5	700	.93	177	$-.032$	27	$+.011$	$-.008$	$+.000$
0.6–0.85	300	.92	47	$-.016$	12	$+.008$	$-.002$	$-.001$
1.0	700	.83	211	$-.041$	63	$+.035$	$-.009$	$+.012$
1.4–1.5	700	.58	213	$-.018$	151	$+.038$	$+.003$	$+.032$
2.0	600	.25	90	$+.004$	261	$+.029$	$+.013$	$+.052$

The simulation shows the coalition rule detecting transfer confined to one block, where the binary rule cannot, and using it to improve performance. The improvement is largest where the retained block is the image block, which (1) reads directly, and smaller where it is the surrogate block, only part of which reaches $\hat{\beta}$. By (15) the criterion the two rules share improves under either departure, so it admits the one that costs AUC on the same terms as the one that gains it.

3.6 Assessment of ϵ_0

Equation (12) does not pin ϵ_0 down, so what (13) costs relative to the best available choice is a question only this section can answer. The four coalition losses are stored per replicate, so (14) can be replayed at any threshold without refitting. We do that on the 600 replicates of the reference cell, pooling $\mathcal{D}_{\varnothing}$, $\mathcal{D}_{\mathsf{XS}}$ and $\mathcal{D}_{\mathrm{dense}}$ and splitting them by the realised $\|\delta_0\|_1$: 300 replicates over six cells at a radius or less, 100 over two cells at two radii or more, and the four cells in between left out.

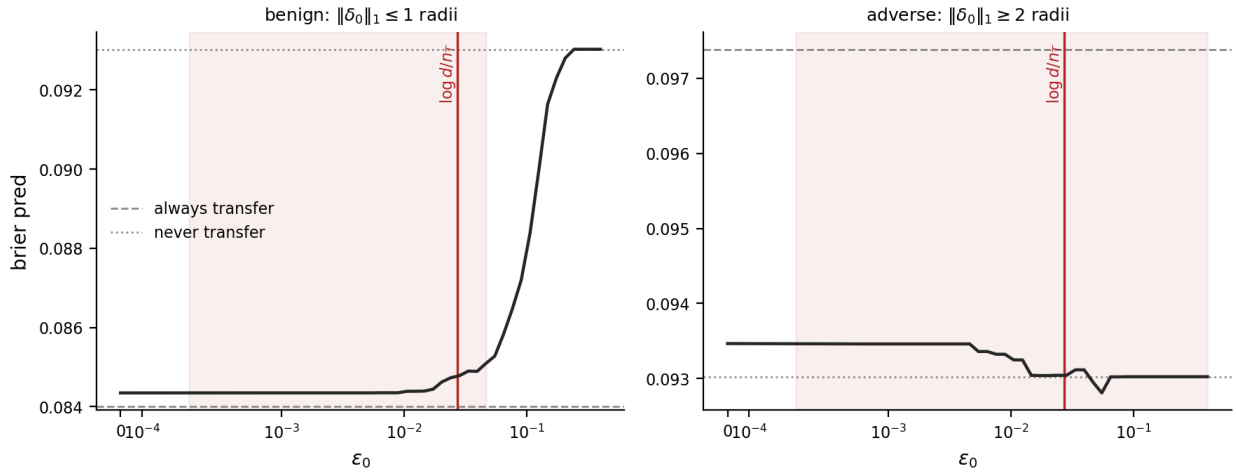


Figure 1: Deployed-layer Brier score of the gate against ϵ_0. The shaded band is where the curve is within 1% of its best. Note the difference in vertical scale between the panels.

Table 10: Excess over the best attainable Brier score at each threshold, in per cent, for the two groups of Figure 1.

ϵ_0	0	.006	.010	**.017**	**.027**	.046	.076	.125
benign	0	.01	.05	.11	**.52**	.89	2.52	6.66
adverse	.71	.59	.47	.25	**.25**	.16	.23	.23
worse of the two	.71	.59	.47	**.25**	**.52**	.89	2.52	6.66

The two curves are optimised in opposite directions. Under a benign defect any threshold that keeps transferring is best, and the excess grows with ϵ_0; under an adverse one the excess falls as ϵ_0 rises and refusals start. A single threshold has to serve both, so what matters is the worse of the two, and that stays within 1% of the best over $\epsilon_0 \in [0, 0.046]$. The minimax point is 0.017 at 0.25%, against 0.52% at $\log d/n_T = 0.027$, the two thresholds a factor of 1.6 apart. That is the order of the multiplier (13) fixes at one, so within the range that matters the rate-based choice is not far from the best a constant could buy.

The RSC tolerance term at the realised norm. Section 2.6 took (12) for its scaling and claimed nothing about its constant. Here $\bar{\theta}$ is known, so the norm the condition is stated in can be evaluated. Write

$$T_{\mathrm{RSC}} \;=\; \frac{\log d}{n_T}\big\|\hat{\theta}^{(\mathrm{meta})} - \bar{\theta}\big\|_1^2, \tag{23}$$

the unit-weight proxy for the RSC tolerance component of (12). Table 11 reports it along the sweep of Section 3.4, where $\|\delta_0\|_1$ is held at 6.09 and only n_T moves.

The norm falls by a fifth across an eightfold increase in n_T, where an $n_T^{-1/2}$ rate would give 65%. It is stable, at eleven to thirteen rather than at the unit scale (13) would need, and stays in that range across the other runs. The factor

Table 11: T_{RSC} at the realised median norm, pooled over $\mathcal{D}_{\mathsf{X}}$ and $\mathcal{D}_{\mathsf{S}}$ at one radius. By (13) and (23) the last column is the square of the third.

n_T	ϵ_0	$\lVert\hat{\theta}^{(\mathrm{meta})} - \bar{\theta}\rVert_1$	T_{RSC}	$T_{\mathrm{RSC}}/\epsilon_0$
100	.0540	13.45	9.76	181
200	.0270	12.58	4.27	158
300	.0180	12.02	2.60	144
400	.0135	11.54	1.80	133
500	.0108	11.26	1.37	127
800	.0067	10.91	0.80	119

multiplying $\log d/n_T$ in (12) is therefore close to constant, which is what (13) needs, but far from the unit constant that would make (13) satisfy the condition: T_{RSC} runs 119 to 181 times ϵ_0, and above $L(\varnothing) = 0.334$ throughout, so no fixed constant reconciles the two here.

That gap does not reach the results. (15) holds identically and without assumptions, and the cost of a larger ϵ_0 is bounded directly by Table 10, which is flat well past the value used.

3.7 What ϕ measures

ϕ measures what Section 2.7 defines it to measure, the marginal value of a block in the imputation fit on the log loss of (11); what follows is what that leaves it unable to do.

ϕ is not a measure of transferability. Section 3.3 has already shown this. Since the gap $\phi_{\mathsf{S}} - \phi_{\mathsf{X}}$ carries a strength term unrelated to δ_0 and the identification criterion reads only the sign of the sum, the ends of the ladder return what a rule naming the weaker block regardless of δ_0 would return, and only near the balanced rung does the rate respond to where δ_0 actually sits.

ϕ is not the value of a block for deployed prediction. At the imputation layer X contributes 14% more than S. At the deployed layer the surrogate block's contribution is indistinguishable from zero while the image block's is not, and the XS row of Table 12 says the same from the other side: its deployed gain is that of X alone. Adding S moves $L(\varnothing) - L(c)$ from $+.0513$ to $+.1152$ and the deployed AUC by $-.0003$ $[-.0034, +.0027]$.

Table 12: The four coalitions under $\mathcal{D}_{\varnothing}$, over 50 paired replicates, each against $\varnothing$ with a 95% interval.

		Deployed AUC gain	
c	$L(\varnothing) - L(c)$	fitted labels	oracle labels
$\varnothing$	—	—	—
X	$+.0513$ $[+.0401, +.0625]$	$+.1493$ $[+.1212, +.1775]$	$-.0004$ $[-.0011, +.0004]$
S	$+.0450$ $[+.0356, +.0544]$	$+.0086$ $[-.0047, +.0219]$	$-.0002$ $[-.0007, +.0003]$
XS	$+.1152$ $[+.1018, +.1286]$	$+.1490$ $[+.1209, +.1771]$	$+.0004$ $[-.0008, +.0016]$

The oracle ablation shows where the surrogate block's contribution goes. Replacing the soft labels of (19) with $\bar{m}(Z)$ of (5) leaves the imputation layer bit-identical, the two runs being paired and the four coalition losses agreeing to machine precision. It collapses the deployed layer: the three point estimates lie within $.0004$ of zero, every interval covers it, and the four coalitions spread by $.0008$ in AUC. A coalition reaches $\hat{\beta}$ through the imputed labels and through no other channel, and of the two blocks only θ_{X} has a counterpart in β (Sections 2.1 and 2.7). What that channel is worth is not small: borrowing lifts the deployed AUC from .6505 to .7995 against an oracle ceiling of .8291, so it recovers 83% of what perfect labels would give.

ϕ does not order coalitions within a design point. Inside a cell the design is held fixed and only the sample changes, so a correlation there asks whether the replicate on which ϕ favours S most is also the replicate on which S actually does best. At the imputation layer, the layer ϕ is defined on, the two do not line up: the median correlations are within .03 of zero, and the cells reaching $p < 0.05$ are 7 and 3 of 76 against the 3.8 chance would give (Table 13). The deployed Brier column is the same. Sign agreement is the wrong column to read here: where the attribution gap sits far from zero the two gaps agree in sign whatever ϕ is doing within the cell, and the ladder of Section 3.3 contains cells at 1.00 agreement with a correlation of $-.38$. The one column that departs from noise is the deployed AUC, at $-.146$ and

negative in 79% of cells; that is the channel above rather than a property of the decomposition. Across cells ϕ moves as it should, the gap turning over with $\|\delta_0\|_1$ in Table 6.

Table 13: Spearman correlations within a design cell, over the 76 cells of the sweeps. Sign agreement is the share of a cell's 50 replicates on which the two gaps agree in sign, averaged over cells.

Layer	Metric	Sign agreement	Median ρ_s	Significant
imputation	AUC	.71	$+.008$	$7/76$
imputation	Brier	.70	$+.021$	$3/76$
deployed	AUC	.51	$-.146$	$19/76$
deployed	Brier	.61	$+.002$	$2/76$

The use ϕ does not support is therefore the one an application has, where there is a single sample and the question is which coalition to take this time.

4 Application

4.1 Data and study design

We use CheXpert-Plus [11, 12], which links each CheXpert radiograph [13] to its de-identified report, split into sections, and to demographic and administrative fields.

The endpoint is pneumothorax. We take labels from the original CheXpert training file rather than from the label columns shipped with CheXpert-Plus, and map them positive-versus-rest: canonical 1 is a case, and 0, -1 and blank are controls. Each row keeps its unmodified canonical state.

The narrative we use as S_{text} is the History and Clinical History sections, written before the radiograph is read; the Findings and Impression sections record the read itself and are excluded. They record the workup: a request to rule out a pneumothorax says someone had already suspected one, a mention of a chest tube that the patient is already being treated for one. This is the clinical context (2) asks S to supply and the image does not carry.

An image enters the cohort if it is a frontal view and if that text is nonempty after cleaning.

Domains are defined by the race field: White is the source domain, Asian the target. It enters the study nowhere else: X and S exclude it, along with ethnicity, vital status and any split indicator, and data preparation asserts this. Classifiers on chest radiographs show subgroup performance gaps [15, 18], and race is recoverable from the pixels alone [17, 16], so the two domains are separable from the image block before any surrogate enters.

Source rows are split once, at the patient level. We draw $n_{\mathcal{S}} = 1200$ distinct source patients and keep one image from each, forming SL; the remaining images of those patients are discarded. All images of the remaining source patients form SU (82,433 images from 30,789 patients).

The target domain contains $16{,}809$ images from $6{,}476$ patients. Where the source split is made once, the target sample is redrawn per repeat. From target patients with at most six images we draw TL and a held-out set VALID of 200 images, one image per patient and patient-disjoint, each stratified to the cohort event rate of 0.105 by fixed count: VALID carries 21 cases in every repeat and TL the 21 or 42 its size implies. The six-image cap applies to both, since patients with many films are inpatients and so at elevated pneumothorax risk. Every image of every unselected target patient forms TU, which carries no per-patient cap.

We run two arms, $n_T = 200$ and $n_T = 400$, at 20 repeats each, and summarise every contrast by its mean over repeats with a paired t-interval, reporting the arms separately. They differ only in n_T, except that the SUPERVISED fine-tuning schedule of Section 4.2 reads it, so an arm difference on that branch is not n_T alone.

Within a repeat the split is fixed before anything is fitted, so the three image branches and all six endpoints of Table 1 see the same TL, VALID and TU. TU is read by the density ratio, the representation, the standardisation and the imputation, so figures computed on it are transductive and reported separately.

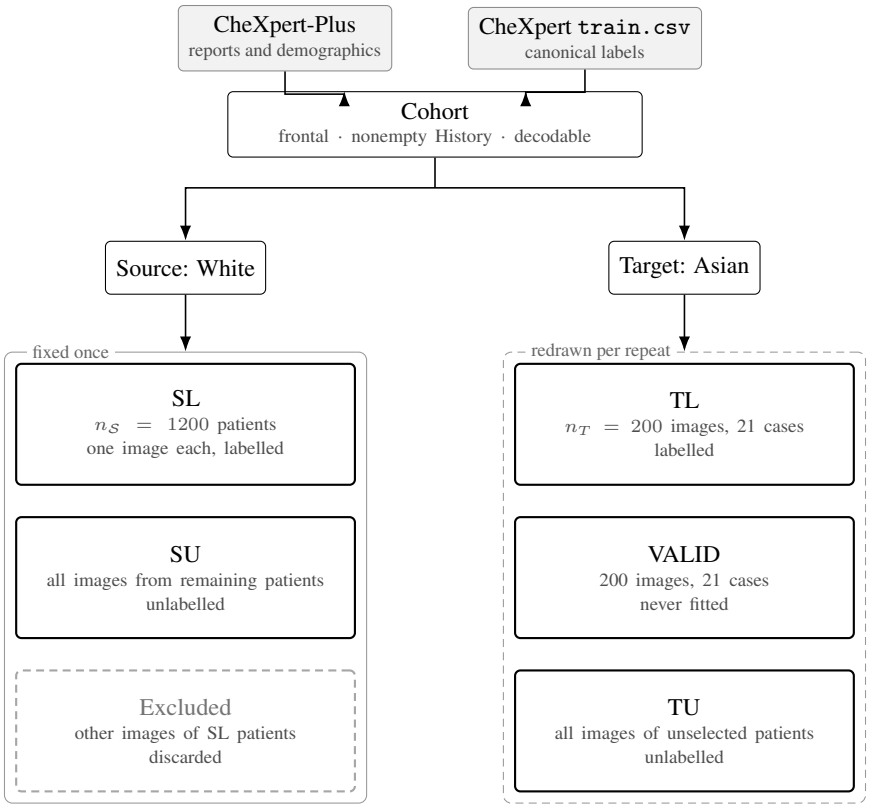


Figure 2: Cohort construction and splits.

4.2 Image representations, surrogates, and comparators

Image representations. The image block comes from an encoder, and we use the three of Table 14. Everything downstream is held fixed, so a difference between branches is attributable to the representation.

Table 14: The three image branches.

Branch	Encoder	Fitted on	Reads Y	Raw dim.	p
BIOVIL	frozen BioViL	—	no	128	37
SIMCLR	DenseNet-121	TL ∪ TU	no	1024	128
SUPERVISED	DenseNet-121	SL, TL	yes	1024	128

BIOVIL. The first branch uses the BioViL image encoder [22], a ResNet-50 trained on MIMIC-CXR: first with SimCLR on the images, then jointly with a radiology-specific text encoder against the paired reports. We take the L_2-normalised 128-dimensional projected embedding and fit nothing.

SIMCLR. The second branch trains a SimCLR encoder [19] on TL ∪ TU: target images, no labels, VALID excluded. The objective is the one BioViL used to pretrain its own backbone, so the two branches share the pretraining algorithm and differ in corpus, in backbone, and in the report supervision layered on top. The backbone is a DenseNet-121 [20] from ImageNet [21], and we keep the 1024-dimensional pooled features.

SUPERVISED. The third branch fine-tunes the same ImageNet-initialised DenseNet-121 on SL and TL under the outcome, with source rows carrying a density ratio estimated in pixel space. That ratio comes from a low-rank projection of the raw images followed by the domain classifier of Section 2.3, without the hard threshold, since nothing downstream reads these coefficients.

It is included because it is what a practitioner with these data would build. Labels enter the representation before they enter the estimator, so this branch is excluded from the permuted-label control; and the fitted network is itself a predictor, which we report as CNN.

From features to X. All three branches end the same way. Features are standardised and reduced by whitened principal components, both fitted on $\mathrm{SL} \cup \mathrm{SU} \cup \mathrm{TL} \cup \mathrm{TU}$, with VALID and the discarded sibling images excluded, and neither reading an outcome. Whitening puts the image block on the unit variance that the threshold κ of Section 2.3 assumes, at the cost of rescaling near-null directions to the same scale as real ones, so we keep only components above 10^{-3} of the leading eigenvalue. SimCLR and Supervised branches keep all 128; BIOVIL keeps 37, which is what its projection head spans. Architecture, optimiser, schedule and augmentation parameters are given in Supplementary Section S2.2.

Surrogate construction. *Structured fields.* The first family is age (standardised, with a missingness indicator and thresholds at 65 and 75), body mass index (the same treatment, thresholds at 30 and 35), sex, whether an interpreter was required, and payer category. The last three enter as indicators with an explicit unknown level. Sixteen columns are built, two are constant in this cohort and dropped, and $q_{\mathrm{meta}} = 14$ remain.

Narrative terms. The second family is term counts from the text of Section 4.1. We count unigrams and bigrams appearing in at least ten documents and take $\log(1 + \cdot)$. Three rules then cut the candidate set, none of them reading an outcome. The complete text processing and vocabulary-filtering rules are given in Supplementary Section S2.3.

Named coordinates. Nothing rotates or reduces the surrogate block, so a coefficient in θ_{S} can be read back to the term it sits on, and an audit file says which rule removed each discarded candidate.

Dimensions. The vocabulary is fixed once on the pooled cohort from document frequencies, so it is transductive but label-free. Both families are then standardised per repeat on $\mathrm{SL} \cup \mathrm{SU} \cup \mathrm{TL} \cup \mathrm{TU}$, with VALID excluded. This leaves $q_{\mathrm{text}} = 79$, so $q = 93$.

Comparators. Supplementary Table S10 lists five estimators outside the construction, fitted on the same features, splits and evaluation rows. SUP is the supervised target-only fit, CS the density-ratio-weighted covariate-shift estimator, and SUPTRANS the three-step transferring procedure of Li et al. [8]; all three are benchmarks of Cai et al. [5, Sec. 4]. CEILING is a ridge logistic fit on TU with its labels observed, and CNN a fine-tuned convolutional network. Neither is a competitor, so both are reported as levels and excluded from the paired contrasts, and CEILING is not scored on TU, for which it is in-sample.

4.3 Results

Diagnostics and scope of the attribution. Supplementary Table S13 collects the nuisance diagnostics; none of them depends on ϵ_0. Every density-ratio fit clears the prespecified separability threshold: the apparent domain AUC is .750 on BIOVIL and .830–.854 on the other two branches. Reweighting brings the worst-coordinate imbalance from 1.006 pooled standard deviations to .504–.627, while the weights retain an effective .26–.38 of $n_{\mathcal{S}} = 1200$ source rows. A null correction says no correctable bias was detected at the selected penalty, not that none is present; its frequency ranges from .60 to .95. The efficiency residual of Proposition 1 is zero to machine precision in all 120 branch-repeat combinations.

Supplementary Tables S14 and S15 locate the three branches on the strength ladder of Section 3.3. Both the target and source rows have to be balanced, and only BIOVIL at $n_T = 200$ qualifies. We therefore use this branch–arm combination to interpret the ordering of ϕ_{X} and ϕ_{S} at the design level. This restriction concerns only the split between the blocks: by Proposition 1, the two attributions sum to $v(\mathsf{XS})$ on every branch.

Predictive performance. Table 15 gives the paired AUC contrasts against TARGET-ONLY on VALID. GATE-BLOCK gains over GATE-BINARY where the two blocks come apart. SIMCLR is the extreme case, with BORROW-S carrying the transfer in both arms while BORROW-X is useless at $n_T = 200$ and harmful at 400; there GATE-BLOCK improves on GATE-BINARY in both, and is the only one of the two to clear zero in the first arm and stay positive in the second. On SUPERVISED the two stay within .002, the sign of the difference turning from negative to positive between the arms. On BIOVIL they are nearly identical at $n_T = 200$, and GATE-BINARY leads by .005 at 400 through the single narrowing discussed below. Both gates stay within .012 AUC of CORRECTED-META across the six branch–arm combinations; that gating costs so little here is an observation, since the gates read log loss, which need not preserve deployed AUC (Section 3.2).

The external comparators sit below the rule. SUP, which discards the unlabelled target rows, trails TARGET-ONLY by .041 to .054 AUC at $n_T = 200$. CS and SUPTRANS match it on BIOVIL and are null or negative elsewhere; both transfer whole, and neither can decline one block while keeping the other.

Brier score and ECE are in Supplementary Tables S20 and S21, and PR-AUC in Supplementary Table S22. On SUPERVISED every construction endpoint except BORROW-S has adverse ECE in both arms, with an interval clear of

Table 15: Paired AUC contrasts against TARGET-ONLY on VALID. PR-AUC, Brier, ECE, levels and transductive results are in Supplementary Section S2.9.

Endpoint	BIOVIL	SIMCLR	SUPERVISED
Panel A: $n_T = 200$			
BORROW-X	$+.022$ $[+.005, +.039]$	$-.001$ $[-.042, +.040]$	$+.004$ $[-.019, +.026]$
BORROW-S	$+.016$ $[+.003, +.029]$	$+.043$ $[+.016, +.070]$	$+.016$ $[+.005, +.028]$
CORRECTED-META	$+.033$ $[+.017, +.049]$	$+.033$ $[-.006, +.072]$	$+.015$ $[-.003, +.033]$
GATE-BINARY	$+.032$ $[+.017, +.048]$	$+.033$ $[-.003, +.069]$	$+.010$ $[-.008, +.027]$
GATE-BLOCK	$+.032$ $[+.017, +.047]$	$+.037$ $[+.006, +.068]$	$+.008$ $[-.009, +.026]$
SUP	$-.050$ $[-.069, -.032]$	$-.054$ $[-.094, -.015]$	$-.041$ $[-.072, -.010]$
CS	$+.030$ $[+.012, +.048]$	$+.006$ $[-.037, +.050]$	$-.006$ $[-.032, +.021]$
SUPTRANS	$+.023$ $[+.005, +.042]$	$-.004$ $[-.042, +.034]$	$+.004$ $[-.021, +.030]$
Panel B: $n_T = 400$			
BORROW-X	$+.022$ $[+.005, +.039]$	$-.023$ $[-.045, -.001]$	$+.012$ $[-.009, +.032]$
BORROW-S	$-.001$ $[-.007, +.005]$	$+.013$ $[-.001, +.026]$	$+.008$ $[-.000, +.015]$
CORRECTED-META	$+.024$ $[+.006, +.041]$	$-.008$ $[-.031, +.014]$	$+.019$ $[+.001, +.038]$
GATE-BINARY	$+.024$ $[+.006, +.041]$	$-.005$ $[-.026, +.016]$	$+.010$ $[-.002, +.022]$
GATE-BLOCK	$+.019$ $[+.002, +.035]$	$+.004$ $[-.014, +.022]$	$+.012$ $[+.000, +.025]$
SUP	$-.008$ $[-.022, +.005]$	$-.056$ $[-.087, -.025]$	$-.030$ $[-.050, -.010]$
CS	$+.021$ $[+.005, +.037]$	$-.031$ $[-.061, -.001]$	$-.007$ $[-.035, +.021]$
SUPTRANS	$+.012$ $[-.004, +.028]$	$-.016$ $[-.047, +.014]$	$+.007$ $[-.016, +.030]$

zero everywhere but BORROW-X at $n_T = 400$, while its AUC contrasts are positive and its Brier contrasts null. The endpoints that suffer are the ones taking θ_{X} from the meta fit; BORROW-S, which keeps it from the target-only fit, is the one that does not, and neither of the other two branches shows the pattern. SUPERVISED is also the only branch whose encoder reads SL and TL labels (Section 4.2). Calibration is the only one of the three metrics that registers any of this, which is the separation Section 2.9 reports three of them to catch. GATE-BLOCK is not exempt: it inherits the calibration of whichever coalition it retains.

Blockwise transfer and attribution. The same four cross-fitted losses give both the attribution and the gate decisions, and the two are usable over different ranges.

The synergy $v(\mathsf{XS}) - v(\mathsf{X}) - v(\mathsf{S})$ is negative on every branch and arm (Supplementary Tables S16 and S17): the narrative surrogates and the image carry partly the same information about the same finding. That redundancy is what leaves a single-block coalition competitive with XS, and is why the coalition rule has anything to decide here that the binary rule does not.

Table 16: Blockwise value and the consequences of gate disagreements. Panel (a) is $n_T = 200$; $\Delta\mathrm{AUC}_{\mathsf{X}}$ and $\Delta\mathrm{AUC}_{\mathsf{S}}$ are BORROW-X and BORROW-S against TARGET-ONLY. Panel (b) pools departures across branches within an arm; criterion gain is $L(c_{\mathrm{binary}}) - L(\hat{c})$ and its AUC column is GATE-BLOCK$-$GATE-BINARY. Complete intervals and branchwise results are in Supplementary Tables S17–S19.

(a) Block values and deployed AUC, $n_T = 200$

Branch	$v(\mathsf{X})$	$v(\mathsf{S})$	$v(\mathsf{XS})$	$\phi_{\mathsf{S}} - \phi_{\mathsf{X}}$	$\Delta\mathrm{AUC}_{\mathsf{X}}$	$\Delta\mathrm{AUC}_{\mathsf{S}}$
BIOVIL	$+.0178$	$+.0457$	$+.0594$	$+.0278$	$+.0224$	$+.0160$
SIMCLR	$-.0031$	$+.0417$	$+.0371$	$+.0448$	$-.0008$	$+.0430$
SUPERVISED	$+.0364$	$+.0244$	$+.0578$	$-.0120$	$+.0038$	$+.0162$

(b) What departures from the binary rule bought

n_T	Departure	n	Criterion gain	Mean ΔAUC	$\Delta\mathrm{AUC} > 0$
200	narrow	9	$+.0117$	$-.0038$	$5/9$
	rescue	4	$+.0367$	$+.0215$	$3/4$
400	narrow	7	$+.0058$	$-.0088$	$2/7$
	rescue	7	$+.0218$	$+.0255$	$7/7$

As an attribution, the strength diagnostic licenses a reading only for BIOVIL at $n_T = 200$, where ϕ_{S} exceeds ϕ_{X} in 16 of 20 repeats with an interval clear of zero: the defect, if there is one, sits in the image block. The four reversals are at the rate the balanced rung of Table 5 misidentifies at, though there across replicates rather than within one design point. Panel (a) reverses that ordering at the deployed layer on two of the three branches, so it does not predict which block will pay in AUC. One of the two is SUPERVISED, whose ϕ is in-sample (Section 2.5).

As a decision rule the range is wider, but the decisions have to be shown insensitive to the threshold first. Replaying (14) across a grid (Supplementary Table S12) finds no cliff at the applied value on any branch: acceptance collapses only at one and a half to four times it. BIOVIL accepts in all 20 repeats from $\epsilon_0 = 0$ out to .015 at $n_T = 400$, against an applied .0122; on the other two branches the counts fall gradually rather than flatly, so that halving the applied threshold or raising it by a fifth moves either rule's count by at most three repeats in twenty.

The two gates part company when the block rule keeps one block rather than both or neither, mainly on SIMCLR. By (15), every departure is no worse on the cross-fitted criterion than the coalition selected by the binary rule, but Panel (b) shows that the two types behave differently at the deployed layer. Rescue improves AUC on average in both arms and in every instance at $n_T = 400$; narrowing does neither, and the one on BIOVIL at $n_T = 400$ buys $+.00008$ on the criterion while giving up .101 AUC, which is by itself the whole of the margin by which GATE-BLOCK trails GATE-BINARY there. Keeping one admitted block pays where the binary rule refuses outright, and carries no protection where it does not.

Negative-control analysis. Two things could produce the rescues above: transferable structure in the source, or the mechanics of choosing from four candidates rather than two. We separate them by permuting the source labels within SL, which breaks $p^{(\mathcal{S})}_{Y|Z}$ as completely as the design allows while leaving every candidate set, split and label-free nuisance quantity in place. SUPERVISED is excluded because its encoder reads those labels.

Table 17: Negative control at the applied ϵ_0, over 20 paired repeats. Block acceptance means $\hat{c} \neq \varnothing$. The p-values are exact McNemar comparisons between real and permuted acceptance for each rule. The final two columns are mean AUC contrasts against TARGET-ONLY under permutation; all eight paired 95% intervals cover zero. Full results are in Supplementary Tables S27 and S28.

		Real acceptances		Permuted acceptances		McNemar p		Permuted ΔAUC	
n_T	Branch	Binary	Block	Binary	Block	Binary	Block	Binary	Block
200	BIOVIL	18	18	2	2	3.1×10^{-5}	3.1×10^{-5}	$-.0011$	$-.0011$
	SIMCLR	14	18	4	4	.0020	1.2×10^{-4}	$-.0051$	$-.0051$
400	BIOVIL	20	20	3	4	1.5×10^{-5}	3.1×10^{-5}	$+.0006$	$+.0014$
	SIMCLR	12	17	3	4	.0039	2.4×10^{-4}	$-.0112$	$-.0099$

Both rules collapse under permutation (Table 17), and the two gaps that matter go with them. On SIMCLR the block rule accepts four and five repeats more than the binary rule with the real source, and zero and one more after permutation; no cell contains a permuted-only acceptance, so each exact McNemar test is a sign test on discordant pairs that all favour the real source. The rescues are therefore not a mechanical consequence of the larger candidate set.

What the control does not deliver is a per-repeat guarantee. Permutation still leaves two to four acceptances in every cell, so an acceptance on any one repeat is not by itself evidence that there was something to borrow. All eight AUC contrasts under permutation cover zero: the residual acceptances buy nothing, and nothing prevents them.

A second gap is not one this control can close. The criterion and the deployed endpoint need not agree within a design point (Section 3.5), so even an acceptance the control would vindicate need not pay on VALID. This is the boundary the departures in Table 16 showed from the other side: the criterion governs what is admitted across repeats, not what any one repeat realises on VALID. The threshold is not calibrated against this control, since that would build its false-acceptance rate into ϵ_0; null-calibrated cutoffs and the threshold replay are in Supplementary Tables S29 and S30.

5 Conclusion and Limitation

5.1 Conclusion

We refine STRIFLE's all-or-nothing transfer decision by resolving the imputation parameter along $(\theta_{\mathsf{X}}, \theta_{\mathsf{S}})$. The comparison between $\varnothing$ and XS becomes a four-coalition problem, and the same cross-fitted losses support both a blockwise gate and a blockwise attribution. We apply the construction to pneumothorax prediction in CheXpert-

Plus, transferring from a White source domain to an Asian target domain under three image representations and two labelled-target budgets.

What the four-coalition rule buys is selective rescue. Where the binary rule rejects the source outright, the blockwise rule can still retain one admitted block. These rescues improve deployed AUC in three of four instances at $n_T = 200$ and in all seven at $n_T = 400$. Two of the seven are SUPERVISED, where the criterion that identified them is in-sample (Section 2.5). Narrowing is different. Where the binary rule has already taken both blocks, cutting to one improves the cross-fitted criterion by construction, but loses AUC on average in both arms. The dominance in (15) holds on that criterion and nowhere else. The dead zone keeps the target-only coalition from being displaced on a thin margin; among coalitions it has already admitted it separates nothing.

Both rules use a dead zone that the existing theory does not make computable in an application (Section 2.6). We adopt $\epsilon_0 = \log d / n_T$ as an operational schedule. In the simulation the applied 0.027 is 1.6 times the empirical minimax point on the replay grid and lies inside the range over which the worse of the two Brier curves stays within 1% of its best; in the application, acceptance collapses only at one and a half to four times the applied values, so none of the prespecified thresholds lies near an observed cliff. What the evidence does not touch is the condition itself: the ℓ_1 error norm decreases only from 13.4 to 10.9 across an eightfold increase in n_T, and the tolerance term evaluated at the realised norm with its constant set to one sits two orders of magnitude above the applied value. A dead zone matching that term would admit no coalition at all.

The attribution is exact, given the four estimated losses. Two blocks need no Monte Carlo approximation, and the imputation-layer gain $v(\mathsf{XS})$ is allocated in full. What it does not deliver is readability. Its ordering tracks baseline block strength as much as transferability. It shows no replicate-level agreement with the two single-block coalitions inside a fixed design point. And it need not carry from the imputation layer to deployed AUC. In the application only BIOVIL at $n_T = 200$ passes the strength diagnostic; the rest of the attribution is descriptive. Coalition losses can still say which block to keep where the ordering cannot say where the defect is.

5.2 Limitations

Dependence on a single branch. The strength diagnostic licenses an attribution ordering only for BIOVIL at $n_T = 200$, so $\phi_{\mathsf{S}} - \phi_{\mathsf{X}}$ cannot be separated from that representation. BIOVIL is a domain-specific encoder whose backbone was trained on MIMIC-CXR jointly against paired reports; it has seen neither CheXpert nor any outcome here, but a representation shaped by clinical text may leave different information in S than one that is not. The reported ordering is conditional on this branch and is not a general ordering of radiographs and clinical history.

The threshold is not verified. $\epsilon_0 = \log d / n_T$ is prespecified and assessed empirically, not derived. It takes the explicit $\log d / n_T$ prefactor of the RSC tolerance component of (12), but the rest of that condition cannot be evaluated in an application (Section 2.6), so whether the applied threshold satisfies it is not answered here; the simulation evidence concerns the schedule's finite-sample behaviour rather than the condition. What survives without (12) is the criterion-level comparison (15); we claim no Theorem 2-type estimation-error guarantee for the schedule.

Differences between representation branches. The three branches differ jointly in their training data and objectives, access to outcome labels, and raw dimension; the fine-tuning schedule of SUPERVISED also changes with n_T. Their differences cannot therefore be assigned to any single property of the representation. The branches are distinct settings in which to examine the construction, not an ablation of encoder characteristics.

Level and scope of inference. The 20 repeats redraw splits from the same cohort rather than sampling independent cohorts. The paired intervals and tests describe aggregate behaviour over these splits, not population sampling uncertainty or the reliability of a decision in an individual repeat. The application covers one retrospective cohort, one endpoint and one transfer direction, and does not establish transportability to other outcomes, populations or domain shifts.

References

[1] Gaël Varoquaux and Veronika Cheplygina. Machine learning for medical imaging: methodological failures and recommendations for the future. *npj Digital Medicine*, 5(1):48, 2022. https://doi.org/10.1038/s41746-022-00592-y.

[2] Cheng Jin, Zhengrui Guo, Yi Lin, Luyang Luo, and Hao Chen. Label-efficient deep learning in medical image analysis: Challenges and future directions. *arXiv preprint arXiv:2303.12484*, 2025. https://arxiv.org/abs/2303.12484.

[3] Tobi Olatunji, Li Yao, Ben Covington, Alexander Rhodes, and Anthony Upton. Caveats in generating medical imaging labels from radiology reports. In *Medical Imaging with Deep Learning (MIDL), Extended Abstract Track*, 2019. https://arxiv.org/abs/1905.02283.

[4] Jue Hou, Zijian Guo, and Tianxi Cai. Surrogate assisted semi-supervised inference for high dimensional risk prediction. *Journal of Machine Learning Research*, 24(265):1–58, 2023. https://www.jmlr.org/papers/v24/21-1075.html.

[5] Tianxi Cai, Mengyan Li, and Molei Liu. Semi-supervised triply robust inductive transfer learning. *Journal of the American Statistical Association*, 120(550):1037–1047, 2025. https://doi.org/10.1080/01621459.2024.2393463.

[6] Sai Li, T. Tony Cai, and Hongzhe Li. Transfer learning for high-dimensional linear regression: Prediction, estimation and minimax optimality. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 84(1):149–173, 2022. https://doi.org/10.1111/rssb.12479.

[7] Ye Tian and Yang Feng. Transfer learning under high-dimensional generalized linear models. *Journal of the American Statistical Association*, 118(544):2684–2697, 2023. https://doi.org/10.1080/01621459.2022.2071278.

[8] Sai Li, Tianxi Cai, and Rui Duan. Targeting underrepresented populations in precision medicine: A federated transfer learning approach. *The Annals of Applied Statistics*, 17(4):2970–2992, 2023. https://doi.org/10.1214/23-AOAS1747.

[9] Lloyd S. Shapley. A value for n-person games. In Harold W. Kuhn and Albert W. Tucker, editors, *Contributions to the Theory of Games, Volume II*, volume 28 of *Annals of Mathematics Studies*, pages 307–317. Princeton University Press, Princeton, NJ, 1953. https://doi.org/10.1515/9781400881970-018.

[10] Aditya Menon and Cheng Soon Ong. Linking losses for density ratio and class-probability estimation. In *Proceedings of the 33rd International Conference on Machine Learning*, volume 48 of *Proceedings of Machine Learning Research*, pages 304–313, 2016. https://proceedings.mlr.press/v48/menon16.html.

[11] Pierre Chambon, Jean-Benoit Delbrouck, Thomas Sounack, Shih-Cheng Huang, Zhihong Chen, Maya Varma, Steven Q. H. Truong, Chu The Chuong, and Curtis P. Langlotz. CheXpert Plus: Augmenting a large chest X-ray dataset with text radiology reports, patient demographics and additional image formats. *arXiv preprint arXiv:2405.19538*, 2024. https://arxiv.org/abs/2405.19538.

[12] Stanford Center for Artificial Intelligence in Medicine and Imaging. CheXpert Plus. Dataset, Stanford University, 2024. https://doi.org/10.71718/6nvz-pm34.

[13] Jeremy Irvin, Pranav Rajpurkar, Michael Ko, Yifan Yu, Silviana Ciurea-Ilcus, Chris Chute, Henrik Marklund, Behzad Haghgoo, Robyn Ball, Katie Shpanskaya, Jayne Seekins, David A. Mong, Safwan S. Halabi, Jesse K. Sandberg, Ricky Jones, David B. Larson, Curtis P. Langlotz, Bhavik N. Patel, Matthew P. Lungren, and Andrew Y. Ng. CheXpert: A large chest radiograph dataset with uncertainty labels and expert comparison. *Proceedings of the AAAI Conference on Artificial Intelligence*, 33(1):590–597, 2019. https://doi.org/10.1609/aaai.v33i01.3301590.

[14] Andrew G. Taylor, Clinton Mielke, and John Mongan. Automated detection of moderate and large pneumothorax on frontal chest X-rays using deep convolutional neural networks: A retrospective study. *PLOS Medicine*, 15(11):e1002697, 2018. https://doi.org/10.1371/journal.pmed.1002697.

[15] Laleh Seyyed-Kalantari, Haoran Zhang, Matthew B. A. McDermott, Irene Y. Chen, and Marzyeh Ghassemi. Underdiagnosis bias of artificial intelligence algorithms applied to chest radiographs in under-served patient populations. *Nature Medicine*, 27(12):2176–2182, 2021. https://doi.org/10.1038/s41591-021-01595-0.

[16] Ben Glocker, Charles Jones, Mélanie Bernhardt, and Stefan Winzeck. Algorithmic encoding of protected characteristics in chest X-ray disease detection models. *eBioMedicine*, 89:104467, 2023. https://doi.org/10.1016/j.ebiom.2023.104467.

[17] Judy W. Gichoya, Imon Banerjee, Ananth Reddy Bhimireddy, John L. Burns, Leo Anthony Celi, Li-Ching Chen, Ramon Correa, Natalie Dullerud, Marzyeh Ghassemi, Shih-Cheng Huang, Po-Chih Kuo, Matthew P. Lungren, Lyle J. Palmer, Brandon J. Price, Saptarshi Purkayastha, Ayis T. Pyrros, Lauren Oakden-Rayner, Chima Okechukwu, Laleh Seyyed-Kalantari, Hari Trivedi, Ryan Wang, Zachary Zaiman, and Haoran Zhang. AI recognition of patient race in medical imaging: a modelling study. *The Lancet Digital Health*, 4(6):e406–e414, 2022. https://doi.org/10.1016/S2589-7500(22)00063-2.

[18] Yuzhe Yang, Haoran Zhang, Judy W. Gichoya, Dina Katabi, and Marzyeh Ghassemi. The limits of fair medical imaging AI in real-world generalization. *Nature Medicine*, 30(10):2838–2848, 2024. https://doi.org/10.1038/s41591-024-03113-4.

[19] Ting Chen, Simon Kornblith, Mohammad Norouzi, and Geoffrey Hinton. A simple framework for contrastive learning of visual representations. In *Proceedings of the 37th International Conference on Machine Learning*, volume 119 of *Proceedings of Machine Learning Research*, pages 1597–1607, 2020. https://proceedings.mlr.press/v119/chen20j.html.

[20] Gao Huang, Zhuang Liu, Laurens van der Maaten, and Kilian Q. Weinberger. Densely connected convolutional networks. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pages 4700–4708, 2017. https://openaccess.thecvf.com/content_cvpr_2017/html/Huang_Densely_Connected_Convolutional_CVPR_2017_paper.html.

[21] Jia Deng, Wei Dong, Richard Socher, Li-Jia Li, Kai Li, and Li Fei-Fei. ImageNet: A large-scale hierarchical image database. In *2009 IEEE Conference on Computer Vision and Pattern Recognition*, pages 248–255, 2009. https://doi.org/10.1109/CVPR.2009.5206848.

[22] Benedikt Boecking, Naoto Usuyama, Shruthi Bannur, Daniel C. Castro, Anton Schwaighofer, Stephanie Hyland, Maria Wetscherek, Tristan Naumann, Aditya Nori, Javier Alvarez-Valle, Hoifung Poon, and Ozan Oktay. Making the most of text semantics to improve biomedical vision–language processing. In *Computer Vision–ECCV 2022*, volume 13696 of *Lecture Notes in Computer Science*, pages 1–21. Springer, 2022. https://doi.org/10.1007/978-3-031-20059-5_1.

[23] Robert Tibshirani. Regression shrinkage and selection via the lasso. *Journal of the Royal Statistical Society: Series B (Methodological)*, 58(1):267–288, 1996. https://doi.org/10.1111/j.2517-6161.1996.tb02080.x.

[24] Amir Beck and Marc Teboulle. A fast iterative shrinkage-thresholding algorithm for linear inverse problems. *SIAM Journal on Imaging Sciences*, 2(1):183–202, 2009. https://doi.org/10.1137/080716542.

[25] Victor Chernozhukov, Denis Chetverikov, Mert Demirer, Esther Duflo, Christian Hansen, Whitney Newey, and James Robins. Double/debiased machine learning for treatment and structural parameters. *The Econometrics Journal*, 21(1):C1–C68, 2018. https://doi.org/10.1111/ectj.12097.

[26] Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q. Weinberger. On calibration of modern neural networks. In *Proceedings of the 34th International Conference on Machine Learning*, volume 70 of *Proceedings of Machine Learning Research*, pages 1321–1330, 2017. https://proceedings.mlr.press/v70/guo17a.html.

[27] Glenn W. Brier. Verification of forecasts expressed in terms of probability. *Monthly Weather Review*, 78(1):1–3, 1950. https://doi.org/10.1175/1520-0493(1950)078%3C0001:VOFEIT%3E2.0.CO;2.

Supplementary Material

for "Selective Blockwise Transfer for Surrogate-Assisted Prediction under Population Shift"

Xiangyu Wang

S1 Simulation Supplement

S1.1 Numerical settings for the simulation

The simulation reimplements Algorithm 1 rather than importing the application pipeline, so that a change in one is not silently inherited by the other. Every constant below is the pipeline's; the one deliberate departure is the solver for the ℓ_1 fits, noted where it arises.

S1.1.1 Reference cell.

$p = 128$ image coordinates, $q_{\text{text}} = 79$ term and $q_{\text{meta}} = 14$ metadata surrogates, $n_T = 200$, $n_{\mathcal{S}} = 1200$, $N_T = N_{\mathcal{S}} = 10^4$, 200 held-out labelled target rows, target prevalence 0.105, AR(1) correlation $\nu = 0.1$, master seed 42. Outcomes are drawn, so the labelled positive count is $\text{Binomial}(n_T, 0.105)$ where the application fixes it at 21 by construction; Section S1.2.1 reports the realised counts.

S1.1.2 The ℓ_1 fits.

$\hat{\theta}^{(\mathcal{T})}$, $\tilde{\theta}$ and $\hat{\beta}$ are fitted by `liblinear` with 5000 iterations, over the fixed grid $C \in \text{logspace}(-3, 1, 9)$ selected by 5-fold stratified cross-validated log loss; the fold count drops to the smaller class count when that is below five, and a fit with fewer than two rows in a class falls back to $C = 1$. The validation loss carries the fitting weights, so for $\tilde{\theta}$ the penalty is selected under a source-weighted risk (Section 2.4). `liblinear` implements the intercept as a regularised synthetic column, whereas the pipeline's `saga` leaves it unpenalised; we set `intercept_scaling` to 10^5, at which the recovered intercept matches the exact unpenalised value to four decimals. At the default of one the difference is not cosmetic: at $C \leq 0.01$ `liblinear` returns an intercept of exactly zero, that is, a model predicting 0.5 for every row at a prevalence of 0.105.

S1.1.3 Density ratio.

Ridge logistic at $C = 1$ by `lbfgs` with 5000 iterations, fitted on the pooled unlabelled rows standardised in place. The hard threshold $\kappa = 10\sqrt{\log d/(N_{\mathcal{S}} + N_T)}$ is applied to the standardised coefficients and the intercept refitted at the survivors; the reported domain AUC is that of the untruncated fit, matching the quantity the application reports under the same name. Weights are centred so that $\hat{E}_{\mathcal{S}}[\hat{w}] = 1$, clipped to $[0.01, 10]$, then renormalised, in that order.

S1.1.4 The correction.

Solved by FISTA [24] with the penalty on slopes only, over the seven ratios $\{1, \frac{1}{2}, \frac{1}{5}, \frac{1}{10}, \frac{1}{20}, \frac{1}{50}, \frac{1}{100}\}$ of $\lambda_{\max}$, the value at which every slope is zero at the refitted intercept, selected by 5-fold stratified cross-validation. The step is $1/L$ with $L = \frac{1}{4}\|A\|_2^2/n$, the cap is 5000 iterations, and the stopping rule is $\|b_{k+1} - b_k\|_\infty \leq 10^{-7}(1 + \|b_k\|_\infty)$. Solves that reach the cap are counted rather than warned about: Section S1.3.4 reports how many do and what re-solving them changes.

S1.1.5 Outcome layer and cross-fitting.

Soft labels enter by row duplication with weights $\hat{m}$ and $1 - \hat{m}$, so each original row carries total weight one and the penalty scale matches an ordinary fit. Folds are stratified on the labelled rows and plain K-fold on the imputed ones, over the same C grid. The coalition losses of (11) use a two-fold stratified split of TL, with the seed offset by fold so that the two folds' internal cross-validation splits are independent.

S1.2 Design noise

S1.2.1 Positive labelled rows

The target prevalence is 0.105 by construction, but the number of positive labelled target rows a replicate draws is not fixed. Counts are shared across the scenarios and magnitudes within a replicate, so Table S1 is over the 50 replicates of a cell. The realised dispersion is binomial: 3.7 to 4.6 at $n_T = 200$ against the 4.3 a binomial draw predicts, and 8.8 at $n_T = 800$ against 8.7.

Table S1: Positive counts by cell, over the 50 replicates of each. The held-out set has 200 rows at every cell.

		Positive labelled target rows			Held-out
Cell	n_T	mean	s.d.	range	positives
reference, both ablations	200	20.5	4.3	10–30	14–28
block strength, 6 rungs	200	20.2–20.6	3.7–4.6	9–33	12–32
n_T sweep	100	10.9	2.9	4–17	14–29
	200	20.5	4.3	10–30	14–28
	300	30.2	4.6	18–39	13–30
	400	43.3	5.5	28–60	14–32
	500	53.1	7.6	36–69	14–32
	800	83.0	8.8	59–98	14–27

S1.2.2 Source prevalence

Δ carries no intercept component, so the source marginal rate moves with the source slopes. Table S2 gives its mean at every magnitude. At two radii the sparse shift has taken it to 0.212 and the dense shift of the same $\|\delta_0\|_1$ only to 0.119.

Table S2: Mean source event rate. The target rate is 0.105 at every cell. The n_T sweep runs at one radius only.

$\|\delta_0\|_1$ (radii)	$\mathcal{D}_{\mathsf{XS}}$	$\mathcal{D}_{\text{dense}}$	6 strength rungs	n_T sweep
0	.107	—	—	—
0.50	.118	.108	.115–.117	—
0.75	.128	—	—	—
1.00	.141	.110	.131–.139	.142–.146
1.25	.156	—	—	—
1.50	.173	.113	.154–.170	—
2.00	.212	.119	.184–.209	—
2.98	.313	—	—	—

S1.3 Numerical checks for the simulation

Section 3.1 asserts four properties of the runs in Table 2. Unless stated otherwise, every count below is over the $4{,}350$ distinct replicates of that table, of which $4{,}250$ carry a non-zero shift.

S1.3.1 Efficiency

The residual $|\phi_{\mathsf{X}} + \phi_{\mathsf{S}} - v(\mathsf{XS})|$ of Proposition 1 is exactly zero in the median replicate and never exceeds 5.6×10^{-17}; in all but one of the runs it never exceeds 2.8×10^{-17}.

S1.3.2 Dominance labels

A replicate is labelled by the block carrying more than 65% of the realised $\|\delta_0\|_1$, and that label never disagrees with the scenario. Under $\mathcal{D}_{\mathsf{X}}$ the image block carries a median 0.853 of $\|\delta_0\|_1$ over 1500 replicates, with a minimum of 0.694 against a cut of 0.65; under $\mathcal{D}_{\mathsf{S}}$ a median 0.145, with a maximum of 0.342 against 0.35. The median image share is 0.509 under $\mathcal{D}_{\mathsf{XS}}$ and 0.500 under $\mathcal{D}_{\text{dense}}$.

S1.3.3 Calibration of $\|\delta_0\|_1$

The scale of Δ is solved at a mean of 3.0 iterations and never more than 7. Of the 4250 perturbed replicates, 18 (0.42%) stop short of the request, seventeen of them $\mathcal{D}_{\mathsf{XS}}$ at three radii where the logit saturates. All 18 fall outside the n_T sweep, whose $\|\delta_0\|_1$ is anchored at the reference cell's radius rather than at its own (Section 3.4); over the remaining cells the realised-to-requested ratio lies in $[0.979, 1.006]$, so none is displaced along its own axis by more than 2.2%.

Table S3: Calibration of $\|\delta_0\|_1$, excluding the n_T sweep of Section 3.4, whose norm is anchored at the reference cell's radius rather than at its own. *Before* and *after* are the realised norm as a fraction of the requested one, under the linearisation of Section 3.1 and after the scale is solved for.

Scenario	$\|\delta_0\|_1$ (radii)	Converged	Iterations	Before	After	$\|\delta_0\|_1/\|\bar{\theta}\|_1$
$\mathcal{D}_{\varnothing}$	0	1.00	0.0	—	—	7×10^{-15}
$\mathcal{D}_{\mathsf{X}}$	0.50	1.00	3.1	1.209	1.002	0.234
	1.00	1.00	2.8	1.276	1.000	0.467
	1.50	1.00	3.0	1.246	1.001	0.701
	2.00	1.00	3.8	1.170	1.002	0.935
$\mathcal{D}_{\mathsf{S}}$	0.50	0.997	3.1	1.208	1.002	0.234
	1.00	1.00	2.7	1.272	1.000	0.467
	1.50	1.00	3.0	1.241	1.001	0.701
	2.00	1.00	3.9	1.163	1.002	0.935
$\mathcal{D}_{\mathsf{XS}}$	0.50	1.00	3.0	1.159	1.001	0.254
	0.75	1.00	2.9	1.196	1.001	0.380
	1.00	1.00	2.6	1.209	1.000	0.507
	1.25	1.00	2.6	1.201	1.001	0.634
	1.50	1.00	3.0	1.180	1.001	0.761
	2.00	1.00	3.7	1.110	1.002	1.016
	3.00	0.66	5.7	0.938	0.994	1.511
$\mathcal{D}_{\mathrm{dense}}$	0.50	1.00	1.1	1.001	1.000	0.253
	1.00	1.00	1.3	0.997	0.998	0.506
	1.50	1.00	1.8	0.990	0.999	0.760
	2.00	1.00	2.0	0.981	0.999	1.013

S1.3.4 The correction solve

Of the 486,402 solves of (9) across the runs of Table 2, 40.0% reach the iteration cap of Section S1.1 (36.7% to 41.9% by run), all at the small-λ end of the path. Reaching the cap is not a failure there: the correction is shrunk to a support the criterion does not select, and the selected λ never falls at the two smallest ratios of the path.

Table S4: The paired re-solve. Each cell is run twice on identical data.

Block strength	Scenarios	Magnitudes	Deployed layer fitted	Pairs
balanced	$\mathcal{D}_{\mathsf{X}}, \mathcal{D}_{\mathsf{S}}$	0.5, 2	no	80
image-dominant	$\mathcal{D}_{\mathsf{X}}, \mathcal{D}_{\mathsf{S}}$	1	no	40
surrogate-dominant	$\mathcal{D}_{\mathsf{X}}, \mathcal{D}_{\mathsf{S}}$	1	no	40
balanced	$\mathcal{D}_{\mathsf{XS}}$	1	yes	12
Total				172

Over the pairs of Table S4, the two arms agree on $\hat{\rho}$ in 172 of 172, on $\hat{c}$ in 172 of 172, and on identification in all 160 pairs where it is defined, at a rate of 0.656 under both. The mean of $\phi_{\mathsf{S}} - \phi_{\mathsf{X}}$ is -0.007526 with 95% interval $[-0.02979, 0.01474]$ against -0.007535 with $[-0.02980, 0.01473]$, and the $\hat{\rho}$ rate is 0.7558 in both. Table S5 gives the absolute differences quantity by quantity.

Table S5: Absolute differences between the two arms over the 172 pairs. $L(\varnothing)$ does not pass through (9), so its invariance confirms that the arms differ only where intended.

Quantity	$\max \lvert\Delta\rvert$	$\mathrm{median}\,\lvert\Delta\rvert$	Pairs with $\lvert\Delta\rvert > 10^{-6}$
$L(\varnothing)$	0	0	0
$L(\mathsf{X})$	8.6×10^{-3}	1.0×10^{-9}	10
$L(\mathsf{S})$	7.0×10^{-3}	3.9×10^{-11}	13
$L(\mathsf{XS})$	1.7×10^{-2}	3.1×10^{-9}	17
ϕ_{X}	9.5×10^{-3}	3.4×10^{-9}	9
ϕ_{S}	7.9×10^{-3}	1.3×10^{-9}	14
$\lVert\hat{\delta}\rVert_1$	3.2×10^{-5}	2.1×10^{-11}	36
$\lVert\hat{\theta}^{(\mathrm{meta})} - \bar{\theta}\rVert_1$	8.6×10^{-5}	2.1×10^{-11}	44
Efficiency residual	2.8×10^{-17}	0	0

The discrepancies concentrate at the larger radius: of the twelve balanced-run pairs whose coalition losses move at all, eight sit at two radii and four at half a radius. The largest single discrepancy is one replicate at $\mathcal{D}_{\mathsf{S}}$ and two radii, where $L(\mathsf{XS})$ falls from 0.3686 to 0.3512 and ϕ_{X} moves from 0.0059 to 0.0154; both arms return $\hat{\rho} = 0$ and $\hat{c} = \varnothing$. In the twelve pairs of Table S4 that fit (19), no AUC differs, the Brier score differs by at most 3.6×10^{-7} and the calibration error by at most 8.5×10^{-7}, and λ_β and $\lVert\hat{\beta}\rVert_0$ are identical in every pair and coalition.

S1.4 Additional simulation results

Table S6 gives both layers along the axis of Section 3.2, for $\mathcal{D}_{\mathsf{XS}}$ and $\mathcal{D}_{\mathrm{dense}}$ and for the endpoints TARGET-ONLY, CORRECTED-META and GATE-BLOCK. The crossings of Table 4 are read from it.

Table S7 resolves the comparison of Section 3.5 by the rungs of Section 3.3. The rise in the deployed difference with the radius reported there is not an artefact of pooling across the ladder: it appears within each rung separately, with rank correlations of $+.12$ to $+.20$ and $p \leq .013$ throughout. What a rung changes is the composition. As the surrogate block gains on the image block, a larger share of the coalition rule's single-block commitments name S, and the deployed difference falls along the same order while the imputation difference does not.

Table S6: Both layers along the axis of Section 3.2, for the endpoints TARGET-ONLY, CORRECTED-META and GATE-BLOCK.

$\|\delta_0\|_1$	Brier			AUC			ECE		
(radii)	T-only	C-meta	Gate	T-only	C-meta	Gate	T-only	C-meta	Gate
Imputation layer, $\theta^{(c)}$ under $\mathcal{D}_{\mathsf{XS}}$									
0	.0895	.0637	.0644	.706	.907	.897	.0452	.0441	.0438
0.50	.0895	.0658	.0679	.706	.901	.876	.0452	.0475	.0465
0.75	.0895	.0701	.0740	.706	.889	.841	.0452	.0496	.0480
1.00	.0895	.0748	.0776	.706	.873	.827	.0452	.0540	.0496
1.25	.0895	.0787	.0837	.706	.854	.783	.0452	.0584	.0516
1.50	.0895	.0859	.0891	.706	.830	.736	.0452	.0644	.0515
2.00	.0895	.0942	.0896	.706	.788	.709	.0452	.0748	.0483
3.00	.0895	.1105	.0897	.706	.698	.705	.0452	.0944	.0460
Deployed layer, $\hat{\beta}(\theta^{(c)})$ under $\mathcal{D}_{\mathsf{XS}}$									
0	.0930	.0831	.0832	.650	.799	.794	.0361	.0410	.0406
0.50	.0930	.0838	.0847	.650	.797	.783	.0361	.0427	.0421
0.75	.0930	.0847	.0864	.650	.783	.751	.0361	.0421	.0406
1.00	.0930	.0865	.0876	.650	.768	.741	.0361	.0447	.0424
1.25	.0930	.0881	.0905	.650	.756	.701	.0361	.0470	.0431
1.50	.0930	.0901	.0926	.650	.741	.671	.0361	.0494	.0406
2.00	.0930	.0935	.0931	.650	.708	.661	.0361	.0540	.0407
3.00	.0930	.1013	.0930	.650	.644	.654	.0361	.0686	.0370
Imputation layer, $\theta^{(c)}$ under $\mathcal{D}_{\mathrm{dense}}$									
0.50	.0895	.0642	.0650	.706	.910	.900	.0452	.0443	.0438
1.00	.0895	.0649	.0657	.706	.906	.896	.0452	.0474	.0465
1.50	.0895	.0658	.0666	.706	.903	.890	.0452	.0472	.0452
2.00	.0895	.0685	.0705	.706	.897	.876	.0452	.0466	.0471
Deployed layer, $\hat{\beta}(\theta^{(c)})$ under $\mathcal{D}_{\mathrm{dense}}$									
0.50	.0930	.0831	.0835	.650	.799	.789	.0361	.0411	.0411
1.00	.0930	.0829	.0833	.650	.797	.787	.0361	.0414	.0409
1.50	.0930	.0832	.0837	.650	.794	.783	.0361	.0412	.0403
2.00	.0930	.0837	.0846	.650	.790	.778	.0361	.0406	.0414

Table S7: The comparison of Section 3.5 by rung. *Share* S is the fraction of the rung's single-block commitments that name the surrogate block. The ΔAUC columns are GATE-BLOCK minus GATE-BINARY, pooled over all replicates at the rung.

	AUC gap	Share	Pooled ΔAUC	
Rung	$\mathsf{S}-\mathsf{X}$	S	deployed	imputation
image-dominant	-0.344	.00	$+.007$	$+.013$
balanced	-0.004	.52	$+.011$	$+.029$
surrogate, soft	$+0.075$	.72	$+.000$	$+.028$
surrogate, medium	$+0.124$	.87	$-.008$	$+.035$
surrogate, strong	$+0.186$	.95	$-.007$	$+.024$
surrogate-dominant	$+0.331$	1.00	$-.008$	$+.020$

S2 Application Supplement

S2.1 Additional cohort and preprocessing details

We use the downsampled CheXpert release, in which images are 320 pixels wide and vary in height around 390. We resize once to 224×224 by area interpolation and store single-channel `uint8`, except in the BIOVIL branch of Section 4.2, which reads the file as released. We apply no histogram equalisation or intensity normalisation here, because each image branch applies its own.

S2.2 Image-representation implementation details

Common image input. For the two DenseNet branches, each stored 224×224 grayscale image is scaled to $[0, 1]$, repeated over three channels, and normalised by the ImageNet channel means $(0.485, 0.456, 0.406)$ and standard deviations $(0.229, 0.224, 0.225)$. Training augmentation consists of a rotation sampled uniformly from $[-5°, 5°]$, multiplication by a factor sampled uniformly from $[0.9, 1.1]$, and independent Gaussian noise with standard deviation 0.01. Intensities are clipped to $[-3, 3]$ after the latter two operations. We use neither cropping nor horizontal flipping. No augmentation is applied during feature extraction or evaluation.

BIOVIL. We use `hi-ml-multimodal` version $0.2.2$ with model type `BIOVIL` and the checkpoint `biovil_image_resnet50_proj_size_128.pt`. The encoder reads the original CheXpert image through its published preprocessing rather than the stored 224×224 array. We extract the L_2-normalised projected global embedding, giving 128 features before the common transformation described in Section 4.2. The encoder is frozen throughout.

SIMCLR. The backbone is an ImageNet-`IMAGENET1K_V1` initialised DenseNet-121 with its classifier replaced by the identity, giving a 1024-dimensional pooled representation. The projection head is

$$1024 \longrightarrow 512 \longrightarrow 128,$$

with a ReLU activation between the two linear layers and L_2 normalisation of the output. The NT-Xent temperature is 0.07.

The encoder is trained for 50 epochs on $\mathrm{TL} \cup \mathrm{TU}$, with VALID and every sibling image of a selected TL or VALID patient excluded. Each image produces two independently augmented views. We use AdamW with learning rate 3×10^{-5} for the backbone, learning rate 3×10^{-4} for the projection head, weight decay 10^{-4}, and batch size 128. No learning-rate scheduler is used. The checkpoint with the lowest mean training contrastive loss is retained. The projection head is then discarded, and the frozen 1024-dimensional backbone output is extracted in batches of 64.

SUPERVISED. The supervised branch uses the same ImageNet-initialised DenseNet-121, with its classifier replaced by a single linear logit. TL is split in a stratified $80{:}20$ ratio. The first part is pooled with SL for training, while the second part is used only for early stopping.

Source observations carry pixel-space density-ratio weights and target observations carry unit weight. Mini-batches are sampled uniformly, so the density ratio enters only once through the weighted binary cross-entropy loss. No class weights are used. Optimisation uses AdamW with weight decay 10^{-4}, a maximum of 50 epochs, and early stopping after five epochs without an improvement of at least 10^{-5} in target validation loss. No learning-rate scheduler is used.

The n_T-specific settings are given in Table S8. For $n_T = 200$, the initial convolution, initial normalisation, dense blocks 1–3, and transitions 1–3 are frozen. For $n_T = 400$, the initial convolution, initial normalisation, dense blocks 1–2, and transitions 1–2 are frozen. After fine-tuning, the fitted network is retained as the CNN endpoint. A copy with its classifier replaced by the identity supplies the 1024-dimensional SUPERVISED representation.

Table S8: Training parameters for the supervised DenseNet-121 branch. Epochs give the maximum before early stopping.

n_T	Batch size	Learning rate	Epochs	Frozen dense blocks
200	8	5×10^{-5}	50	1–3
400	16	1×10^{-4}	50	1–2

Pixel-space density ratio. For the supervised branch, raw images scaled to $[0, 1]$ are mean-centred and projected onto 24 truncated-SVD components. The SVD is fitted on at most $20{,}000$ images drawn proportionally from SU and

TU. A ridge logistic domain classifier with $C = 1$ is fitted on the standardised projected features. Its coefficients are not hard-thresholded. The resulting source weights are centred to have source mean one, clipped to $[0.01, 10]$, and renormalised to mean one.

S2.3 Surrogate vocabulary construction

Text processing. We normalise Unicode, lowercase, collapse whitespace, and strip the quoting artefacts and underscore runs that de-identification leaves behind. Slashes and hyphens are kept, because `s/p`, `r/o` and `w/o` mean something in this register and a standard tokeniser would break them apart.

Which terms. Three rules then cut the candidate set, none of them reading an outcome.

1. A term must clear a 1% document-frequency floor in both domains.
2. Boilerplate goes: bare numerals, tokens naming age or sex, and terms with no clinical content, age and sex being already in the structured family.
3. A unigram goes when a bigram containing it has subsumed it, co-occurring in at least 0.9 of its documents, since near-duplicate columns let the penalty split a coefficient between them arbitrarily.

Negation and uncertainty markers are exempt from the last two. A bag of words tells an affirmative mention from a ruled-out one only through *no*, *without*, `r/o` and their kin, which the boilerplate rule would take for function words and the subsumption rule would delete, *no* occurring almost only inside the phrase it negates.

The floor is two-sided, and at 1%, because $\hat{\delta}$ is estimated on TL alone: a term rare in the target is absent from those n_T rows often enough that $\hat{\delta}$ has no data with which to remove a coefficient $\tilde{\theta}$ put on it, and a term frequent in one domain only fails the same way while also being what makes the two domains easy to tell apart, the quantity Section 2.3 monitors. Table S9 collects the parameters.

Table S9: Parameters used to construct the narrative surrogate block. All filters are label-free.

Parameter	Value
Text fields	History coalesced with Clinical History
Case	lowercase after Unicode NFKC normalisation
n-gram range	unigrams and bigrams
Initial document-frequency floor	at least 10 documents in the pooled corpus
Domain-specific prevalence floor	at least 0.01 in both source and target
Subsumption threshold	0.9 for removing a unigram represented by a retained bigram
Count transformation	$\log(1 + \text{count})$
Protected terms	negation and uncertainty markers are exempt from boilerplate and subsumption removal
Dimension reduction	none
Final text dimension	$q_{\text{text}} = 79$
Final metadata dimension	$q_{\text{meta}} = 14$
Final surrogate dimension	$q = 93$

S2.4 Estimator tuning and numerical settings

Reproducibility. The global seed is 42. Repeat r, indexed from zero, uses seed

$$42 + 1000r + n_T.$$

We run 20 repeats for each of $n_T = 200$ and $n_T = 400$. PyTorch, NumPy and Python random-number generators receive the same repeat-specific seed, and deterministic cuDNN execution is enabled.

Density-ratio estimation. The STRIFLE density ratio is fitted by ridge logistic domain classification on the standardised pooled SU and TU design, using $C = 1$, the `lbfgs` solver, and at most 5000 iterations. No class weights are used. Standardised slope coefficients below

$$10\sqrt{\frac{\log d}{N_{\mathcal{S}} + N_T}}$$

in absolute value are set to zero. The log ratio evaluated on SL is centred so that its exponential has source mean one. The weights are then clipped to $[0.01, 10]$ and renormalised to mean one.

The image-only density ratio used by CS is fitted in the same way but without hard thresholding. The separate pixel-space density ratio used for the supervised DenseNet is described in Section S2.2.

Penalised logistic fits. The target-only imputation fit, the pooled fit, and the final outcome fit use an L_1-penalised logistic model. Their candidate inverse penalties are

$$\mathcal{C} = \left\{10^{-3+0.5j} : j = 0, \ldots, 8\right\} = \left\{10^{-3}, 10^{-2.5}, \ldots, 10^{1}\right\}.$$

Selection uses five-fold stratified cross-validated log loss. The solver is `saga`, with at most $10{,}000$ iterations and tolerance 10^{-4}. No class weights are used, and intercepts are not penalised.

For the pooled imputation fit, source observations carry the density-ratio weights and target observations carry unit weight. The cross-validation loss uses the same fitting weights rather than being restricted to the target validation rows.

Bias correction. The offset L_1 correction is fitted by FISTA. Its candidate penalties are

$$\lambda_{\max} \left\{1,\ 0.5,\ 0.2,\ 0.1,\ 0.05,\ 0.02,\ 0.01\right\},$$

where $\lambda_{\max}$ is the smallest penalty that sets every correction slope to zero after refitting the unpenalised intercept. The penalty is selected by five-fold stratified cross-validated target log loss. Each FISTA solve uses at most 5000 iterations and tolerance 10^{-7}.

Outcome model. The final image-only outcome model uses the same grid $\mathcal{C}$, solver and five-fold log-loss criterion. Soft-labelled observations are represented by two copies with outcomes one and zero and weights m_i and $1 - m_i$, respectively. Thus every original observation contributes total weight one.

Fixed-penalty comparators. The SUP, CS, and CEILING estimators are ridge logistic regressions with $C = 0.1$, the `lbfgs` solver, and at most 2000 iterations. SUP is fitted on TL; CS is fitted on SL $\cup$ TL, with image-only density-ratio weights on source rows and unit weights on target rows; and CEILING is fitted on the observed labels of TU.

SUPTRANS. The reported SUPTRANS estimator uses the complete three-step procedure. Step 1 fits a source-only L_1-penalised logistic model using $\mathcal{C}$. Step 2 fits an offset L_1 target correction using the correction grid above. Before Step 3, only the

$$k = \left\lfloor \sqrt{\frac{n_T}{\log p}} \right\rfloor$$

largest correction coefficients in absolute value are retained. Step 3 fits a joint penalised model on SL $\cup$ TL, with the retained correction subtracted from the source-row linear predictors. Its penalty is selected from the same $\lambda_{\max}$-relative grid by five-fold cross-validation. The cross-validation strata combine population and outcome, and the validation criterion is the pooled loss rather than target-only loss. The three-step estimator is reported as SUPTRANS; its optional binary aggregation with a target-only fit is stored as a separate endpoint.

Table S10: External comparators. All read X alone except CNN, which reads pixels. Neither density ratio here is the one of Section 2.3, which only the construction uses.

	Fitted on	Density ratio	Form
SUP	TL	—	ridge logistic
CS	SL, TL	image block	ridge logistic
SUPTRANS	SL, TL	—	three-step ℓ_1
CEILING	TU, observed labels	—	ridge logistic
CNN	SL, TL	pixel space	fine-tuned DenseNet-121

S2.5 Gate replay and evaluation parameters

Applied dead zone. The reported gate uses

$$\epsilon_0 = \frac{\log d}{n_T}, \qquad d = p + q,$$

with $q = 93$. The realised branch-specific values are given in Table S11.

Table S11: Branch-specific dead-zone values used in the application.

Branch	p	d	$\epsilon_0, n_T = 200$	$\epsilon_0, n_T = 400$
BIOVIL	37	130	0.0243	0.0122
SIMCLR	128	221	0.0270	0.0135
SUPERVISED	128	221	0.0270	0.0135

All four coalition losses were stored before thresholding. Because ϵ_0 changes only the decision and not any fitted estimator, the reported binary and blockwise gates were reconstructed from those losses without refitting. The original endpoints at $\epsilon_0 = 0$ were reproduced exactly as a numerical check before the reported threshold was applied.

For sensitivity analysis, the stored losses were replayed over

$$\epsilon_0 \in \{0,\ .001,\ .002,\ .003,\ .005,\ .0075,\ .010,\ .0125,\ .015,\\ .020,\ .025,\ .030,\ .040,\ .050,\ .075,\ .100,\ .150,\ .200\}.$$

The null acceptance target of 0.10 is used only as a diagnostic and does not select the reported threshold. Threshold-robust separation is assessed at the 0.05 level.

Threshold sensitivity. The four coalition losses are stored per repeat, so (14) can be replayed at any threshold without refitting. Table S12 counts the decisions across a grid.

Table S12: Decisions of (14) replayed across a grid of thresholds, over 20 repeats. Entries are $\hat{\rho} = 1$ / $\hat{c} \neq \varnothing$. The applied $\epsilon_0 = \log d/n_T$ is .0243 on BIOVIL and .0270 on the other two at $n_T = 200$, and .0122 and .0135 at 400; it falls between grid points in every case.

$n_T = 200$				$n_T = 400$			
ϵ_0	BIOVIL	SIMCLR	SUPER.	ϵ_0	BIOVIL	SIMCLR	SUPER.
.0000	20/20	18/20	18/20	.0000	20/20	18/20	17/19
.0010	20/20	18/20	18/20	.0005	20/20	18/20	17/19
.0020	20/20	18/20	18/20	.0010	20/20	17/20	17/19
.0030	20/20	18/20	18/20	.0015	20/20	17/20	17/19
.0050	20/20	18/20	18/20	.0020	20/20	16/19	17/19
.0075	20/20	18/20	18/19	.0030	20/20	14/19	17/19
.0100	18/19	17/20	18/19	.0040	20/20	14/19	17/19
.0125	18/18	16/19	18/18	.0050	20/20	14/19	17/19
.0150	18/18	16/19	18/18	.0075	20/20	14/19	17/18
.0200	18/18	14/19	17/18	.0100	20/20	13/18	17/18
.0250	18/18	14/19	16/16	.0125	20/20	12/17	16/17
.0300	18/18	13/17	15/15	.0150	20/20	11/17	14/16
.0400	17/17	8/9	14/14	.0200	18/18	11/15	14/15
.0500	8/9	7/7	13/13	.0300	14/14	8/9	14/14
.0750	4/4	1/3	5/6	.0500	7/7	3/3	10/10
.1000	2/2	0/0	3/3	.1000	0/0	0/0	1/1
.1500	1/1	0/0	0/0	.2000	0/0	0/0	0/0
.2000	1/1	0/0	0/0				

Evaluation. Expected calibration error uses ten equal-width probability bins. AUC, PR-AUC and Brier intervals for an individual fitted model use 500 patient-clustered bootstrap replicates. The application tables instead report the mean within-repeat contrast against TARGET-ONLY over the 20 repeats, together with a paired 95% t-interval. VALID is the primary held-out evaluation set. Results on TU are reported separately as transductive because TU enters representation learning, standardisation, density-ratio estimation and soft-label imputation.

S2.6 Nuisance diagnostics

Table S13 collects the quantities Sections 2.3, 2.4 and 2.7 promise, as medians over the 20 repeats of each arm; none of them depends on ϵ_0. Support and ESS are fractions of d and $n_{\mathcal{S}}$; B_∞ and $\bar{B}$, the worst-coordinate and the mean

absolute imbalance of (4), are in pooled standard deviations; λ_θ and λ_δ are multiples of the rates $\sqrt{n^{-1}\log d}$ and $\sqrt{n_T^{-1}\log d}$ assigned to them; and "null" is the share of repeats in which $\hat{\delta}$ corrects the intercept only.

Table S13: Nuisance diagnostics.

						$\bar{B}$			
Branch	n_T	AUC	support	ESS	B_∞	wtd.	unwtd.	λ_θ / λ_δ	null
BIOVIL	200	.750	.18	.38	.504	.066	.076	.12 / .34	.60
	400	.750	.19	.38	.513	.066	.076	.11 / .37	.70
SIMCLR	200	.854	.29	.26	.625	.068	.070	.12 / .37	.80
	400	.853	.28	.27	.627	.068	.070	.11 / .34	.95
SUPERVISED	200	.830	.24	.30	.549	.064	.068	.12 / .31	.75
	400	.834	.24	.29	.574	.067	.070	.03 / .19	.90

The density ratio.

1. *Separability.* A fit is read without qualification only below an apparent domain AUC of 0.95. Every fit clears it: $.750$ on BIOVIL, $.830$ to $.854$ on the other two.
2. *Worst-coordinate imbalance.* Unweighted, the two designs differ by 1.006 pooled standard deviations. Reweighting brings this to $.504$ on BIOVIL, $.549$ to $.574$ on SUPERVISED and $.625$ to $.627$ on SIMCLR.
3. *Mean imbalance.* From $.068$–$.076$ unweighted to $.064$–$.068$ weighted. The gain at the worst coordinate does not extend to the rest.
4. *Effective sample and support.* Of $n_{\mathcal{S}} = 1200$ source rows the weights retain an effective $.26$ to $.38$; of d coefficients the threshold retains $.18$ to $.29$. Clipping touches under 1% of rows.

The imputation model.

1. λ_θ. It comes out at $.03$ to $.12$ of the assigned rate $\sqrt{n^{-1}\log d}$, but the value carries little: only four adjacent points of the nine-point grid are ever selected, and never one at either end.
2. *How often the correction is null.* The path is anchored at the penalty that zeroes every slope, and $\hat{\delta}$ stops there in 60% of BIOVIL repeats at $n_T = 200$, rising to 95% on SIMCLR at 400, correcting the intercept alone.
3. *How large it is otherwise.* A median of one to eighteen coordinates, $\|\hat{\delta}\|_1$ between 0.00 and 1.66, mostly on θ_{X} on SUPERVISED and mostly on θ_{S} on the other two.

A null correction says no correctable bias was detected at the selected penalty, not that none is present (Section 2.4). But the anchoring in item 2 narrows that: the criterion had looser penalties available and declined them. The efficiency residual of Proposition 1 is zero to machine precision in all 120 branch-repeat combinations.

S2.7 Locating the branches on the strength ladder

Section 3.3 left the ordering of ϕ_{X} against ϕ_{S} readable only where the two blocks are comparably strong, and measured how far from that condition the ordering stops meaning anything.

Table S14: Single-block and joint AUC on VALID over 20 repeats, $n_T = 200$. Both rows of a branch are scored on the same VALID rows and differ only in what they were fitted on. The gap is $\mathsf{S} - \mathsf{X}$ with a paired 95% interval, so it is the mean of the paired differences and need not equal the difference of the two rounded columns; the last column places it on the ladder of Table 5. The S column does not vary by branch, since S does not depend on the image encoder.

Branch	Fitted on	X only	S only	Joint	Gap $\mathsf{S} - \mathsf{X}$	Rung
BIOVIL	TL	.652	.629	.684	$-.023$ [$-.066, +.020$]	balanced
	SL	.732	.758	.796	$+.026$ [$+.004, +.047$]	
SIMCLR	TL	.542	.629	.629	$+.087$ [$+.042, +.133$]	soft–medium
	SL	.632	.758	.752	$+.125$ [$+.087, +.164$]	
SUPERVISED	TL	.613	.629	.696	$+.017$ [$-.035, +.068$]	balanced–soft
	SL	.672	.758	.758	$+.086$ [$+.046, +.126$]	

Table S15: The same diagnostic at $n_T = 400$, over 20 repeats. Layout and conventions as in Table S14.

Branch	Fitted on	X only	S only	Joint	Gap $\mathsf{S} - \mathsf{X}$	Rung
BIOVIL	TL	.730	.772	.792	$+.042$ [$+.006, +.078$]	balanced–soft
	SL	.731	.771	.803	$+.040$ [$+.006, +.073$]	
SIMCLR	TL	.599	.772	.749	$+.173$ [$+.138, +.208$]	medium–strong
	SL	.635	.771	.741	$+.136$ [$+.091, +.181$]	
SUPERVISED	TL	.666	.772	.750	$+.106$ [$+.067, +.145$]	soft–medium
	SL	.683	.771	.773	$+.087$ [$+.048, +.127$]	

Both rows have to be balanced, not just the first. The TL row is the target-only half of every coalition; the SL row stands in for $\hat{\theta}^{(\mathrm{meta})}$, which is fitted on SL $\cup$ TL with six source rows to every target row in this arm. Only BIOVIL qualifies: $-.023$ and $+.026$, against $+.087$ and $+.125$ for SIMCLR, and $+.017$ and $+.086$ for SUPERVISED, whose TL row sits at the balanced rung but whose SL row does not. We read the attribution ordering off BIOVIL at $n_T = 200$ and treat the other two branches as consistency checks.

The second arm is reported and not read. Every branch moves towards S there, BIOVIL to $+.042$ and $+.040$, and the source rows outnumber the target rows three to one rather than six, at which point TL has enough say in $\hat{\theta}^{(\mathrm{meta})}$ that the SL row no longer stands in for it.

By Proposition 1 the two attributions sum to $v(\mathsf{XS})$ on every branch, so how much borrowing was worth needs no condition from this section. The split between the blocks does. On BIOVIL at $n_T = 200$ the sign of $\phi_{\mathsf{S}} - \phi_{\mathsf{X}}$ names the block carrying the defect at about the error rate of the balanced rung, and no better, since the repeats resample one cohort rather than draw new samples. Two limits survive the balance. The ordering holds across design points and not within one, by Section 3.7, and this is a single design point. And each ϕ_b sums a bias and a variance term of opposite sign (Section 2.7).

S2.8 Blockwise transfer and attribution

Coalition values and attribution. Table S17 reports the quantities behind the blockwise attribution. The value $v(c) = L(\varnothing) - L(c)$ is defined on the two-fold cross-fitted imputation loss. Synergy is $v(\mathsf{XS}) - v(\mathsf{X}) - v(\mathsf{S})$.

Table S16: The four coalition losses of (11) and the attribution of (17) at $n_T = 200$, over 20 repeats. Losses are means; ϕ carries a 95% paired t-interval. Values are recovered as $v(c) = L(\varnothing) - L(c)$ and are not tabulated separately. The synergy column is $v(\mathsf{XS}) - v(\mathsf{X}) - v(\mathsf{S})$, half of which enters each ϕ by construction; it is negative on every branch, so each block is worth less alongside the other than alone. By Proposition 1 the two ϕ sum to $L(\varnothing) - L(\mathsf{XS})$, and the residual is zero to machine precision in all 60 branch–repeat combinations of this arm. These losses are evaluated at the imputation layer, on held-out folds of TL, and are the criterion the selection rules of (14) read; they are not endpoint performance.

	Coalition loss $L(c)$					Attribution	
Branch	$\varnothing$	X	S	XS	Synergy	ϕ_{X}	ϕ_{S}
BIOVIL	.345	.327	.299	.286	$-.0041$	$+.0158$ [$+.0099$, $+.0217$]	$+.0436$ [$+.0256$, $+.0617$]
SIMCLR	.348	.351	.307	.311	$-.0014$	$-.0038$ [$-.0141$, $+.0065$]	$+.0410$ [$+.0297$, $+.0522$]
SUPERVISED	.299	.263	.274	.241	$-.0029$	$+.0349$ [$+.0212$, $+.0486$]	$+.0229$ [$+.0112$, $+.0346$]

Table S17: Coalition values and block attribution over 20 repeats. Bracketed entries are paired 95% t-intervals. Synergy is reported as a point estimate, and the last column counts repeats satisfying the strict inequality.

(a) Coalition values

Branch	n_T	$L(\varnothing)$	$v(\mathsf{X})$	$v(\mathsf{S})$	$v(\mathsf{XS})$	Synergy
BIOVIL	200	.345	$+.0178$ [$+.0126$, $+.0231$]	$+.0457$ [$+.0285$, $+.0629$]	$+.0594$ [$+.0391$, $+.0798$]	$-.0041$
	400	.319	$+.0179$ [$+.0130$, $+.0227$]	$+.0308$ [$+.0211$, $+.0404$]	$+.0468$ [$+.0361$, $+.0576$]	$-.0018$
SIMCLR	200	.348	$-.0031$ [$-.0138$, $+.0075$]	$+.0417$ [$+.0305$, $+.0528$]	$+.0371$ [$+.0256$, $+.0486$]	$-.0014$
	400	.330	$+.0024$ [$-.0064$, $+.0113$]	$+.0243$ [$+.0175$, $+.0311$]	$+.0247$ [$+.0138$, $+.0356$]	$-.0020$
SUPERVISED	200	.299	$+.0364$ [$+.0214$, $+.0513$]	$+.0244$ [$+.0117$, $+.0371$]	$+.0578$ [$+.0407$, $+.0749$]	$-.0029$
	400	.271	$+.0298$ [$+.0157$, $+.0439$]	$+.0140$ [$+.0039$, $+.0241$]	$+.0435$ [$+.0272$, $+.0597$]	$-.0004$

(b) Attribution

Branch	n_T	ϕ_{X}	ϕ_{S}	$\phi_{\mathsf{S}} - \phi_{\mathsf{X}}$	$\phi_{\mathsf{S}} > \phi_{\mathsf{X}}$
BIOVIL	200	$+.0158$ [$+.0099$, $+.0217$]	$+.0436$ [$+.0256$, $+.0617$]	$+.0278$ [$+.0103$, $+.0454$]	16/20
	400	$+.0170$ [$+.0123$, $+.0216$]	$+.0299$ [$+.0203$, $+.0394$]	$+.0129$ [$+.0024$, $+.0234$]	16/20
SIMCLR	200	$-.0038$ [$-.0141$, $+.0065$]	$+.0410$ [$+.0297$, $+.0522$]	$+.0448$ [$+.0266$, $+.0630$]	19/20
	400	$+.0014$ [$-.0071$, $+.0099$]	$+.0233$ [$+.0168$, $+.0298$]	$+.0218$ [$+.0114$, $+.0323$]	18/20
SUPERVISED	200	$+.0349$ [$+.0212$, $+.0486$]	$+.0229$ [$+.0112$, $+.0346$]	$-.0120$ [$-.0308$, $+.0069$]	7/20
	400	$+.0297$ [$+.0166$, $+.0427$]	$+.0138$ [$+.0047$, $+.0229$]	$-.0159$ [$-.0314$, $-.0003$]	6/20

Gate decisions. Table S18 counts the decisions at the applied threshold. A narrowing has $\hat{\rho} = 1$ and $\hat{c} \in \{\mathsf{X}, \mathsf{S}\}$; a rescue has $\hat{\rho} = 0$ and $\hat{c} \in \{\mathsf{X}, \mathsf{S}\}$.

Table S18: Binary and blockwise decisions at the applied ϵ_0, over 20 repeats per branch and arm.

				Block rule $\hat{c}$				Rules disagree		
Branch	n_T	ϵ_0	$\hat{\rho} = 1$	$\varnothing$	X	S	XS	narrow	rescue	total
BIOVIL	200	.0243	18	2	0	1	17	1	0	1
	400	.0122	20	0	0	1	19	1	0	1
SIMCLR	200	.0270	14	2	0	10	8	6	4	10
	400	.0135	12	3	1	9	7	5	5	10
SUPERVISED	200	.0270	16	4	1	1	14	2	0	2
	400	.0135	14	4	2	1	13	1	2	3

What the departures bought. For the binary rule, let $c_{\text{binary}} = \mathsf{XS}$ when $\hat{\rho} = 1$ and $c_{\text{binary}} = \varnothing$ otherwise. Table S19 compares it with the coalition selected by the block rule.

Table S19: Departures of GATE-BLOCK from GATE-BINARY. Criterion gain is $L(c_{\text{binary}}) - L(\hat{c})$ and is averaged within each type of departure. The endpoint columns are GATE-BLOCK−GATE-BINARY on VALID; positive ΔAUC and negative ΔECE favour the block rule. The all-repeat rows include zeros from repeats on which the rules agree and give paired 95% t-intervals for ΔAUC.

n_T	Branch	Departure	n	Criterion gain	ΔAUC	ΔAUC>0	ΔECE
200	BIOVIL	narrow	1	+.0036	−.0035	0/1	−.0114
		all repeats	20	—	−.0002 [−.0005, +.0002]	0/20	—
	SIMCLR	narrow	6	+.0145	−.0008	4/6	−.0121
		rescue	4	+.0367	+.0215	3/4	−.0116
		all repeats	20	—	+.0041 [−.0156, +.0237]	7/20	—
	SUPERVISED	narrow	2	+.0075	−.0128	1/2	−.0090
		all repeats	20	—	−.0013 [−.0043, +.0018]	1/20	—
400	BIOVIL	narrow	1	+.00008	−.1008	0/1	−.0001
		all repeats	20	—	−.0050 [−.0156, +.0055]	0/20	—
	SIMCLR	narrow	5	+.0062	+.0085	2/5	−.0031
		rescue	5	+.0214	+.0260	5/5	+.0108
		all repeats	20	—	+.0086 [−.0053, +.0225]	7/20	—
	SUPERVISED	narrow	1	+.0095	−.0032	0/1	+.0034
		rescue	2	+.0229	+.0242	2/2	−.0008
		all repeats	20	—	+.0023 [−.0014, +.0059]	2/20	—

S2.9 Predictive performance

Tables S20 and S21 report the paired contrasts on VALID. The five nonbaseline construction endpoints appear above the rule and the three external comparators below it. Contrasts are means over the 20 repeats of an arm with paired 95% t-intervals, taken against TARGET-ONLY; negative ΔBrier and ΔECE are improvements. Table S22 reports PR-AUC over the same repeats, since the cohort event rate of 0.105 leaves AUC insensitive to the positive class.

Table S20: Paired contrasts against TARGET-ONLY on VALID, $n_T = 200$.

Branch	Endpoint	ΔAUC	ΔBrier	ΔECE
BIOVIL	BORROW-X	+.022 [+.005, +.039]	−.0034 [−.0045, −.0023]	−.0124 [−.0228, −.0020]
	BORROW-S	+.016 [+.003, +.029]	−.0013 [−.0021, −.0004]	−.0010 [−.0078, +.0058]
	CORRECTED-META	+.033 [+.017, +.049]	−.0044 [−.0057, −.0032]	−.0098 [−.0201, +.0005]
	GATE-BINARY	+.032 [+.017, +.048]	−.0041 [−.0055, −.0027]	−.0076 [−.0174, +.0022]
	GATE-BLOCK	+.032 [+.017, +.047]	−.0039 [−.0054, −.0025]	−.0082 [−.0179, +.0015]
	SUP	−.050 [−.069, −.032]	+.0025 [+.0004, +.0045]	+.0043 [−.0080, +.0167]
	CS	+.030 [+.012, +.048]	−.0033 [−.0051, −.0015]	−.0029 [−.0132, +.0075]
	SUPTRANS	+.023 [+.005, +.042]	−.0039 [−.0055, −.0024]	−.0123 [−.0220, −.0027]
SIMCLR	BORROW-X	−.001 [−.042, +.040]	+.0001 [−.0021, +.0024]	+.0077 [−.0033, +.0187]
	BORROW-S	+.043 [+.016, +.070]	−.0021 [−.0031, −.0011]	+.0002 [−.0056, +.0061]
	CORRECTED-META	+.033 [−.006, +.072]	−.0021 [−.0038, −.0005]	+.0053 [−.0041, +.0148]
	GATE-BINARY	+.033 [−.003, +.069]	−.0018 [−.0034, −.0002]	+.0063 [−.0027, +.0152]
	GATE-BLOCK	+.037 [+.006, +.068]	−.0018 [−.0031, −.0005]	+.0003 [−.0077, +.0084]
	SUP	−.054 [−.094, −.015]	+.0098 [+.0072, +.0124]	+.0421 [+.0308, +.0535]
	CS	+.006 [−.037, +.050]	+.0055 [+.0023, +.0088]	+.0426 [+.0295, +.0556]
	SUPTRANS	−.004 [−.042, +.034]	−.0012 [−.0029, +.0005]	−.0117 [−.0221, −.0012]
SUPERVISED	BORROW-X	+.004 [−.019, +.026]	+.0008 [−.0014, +.0030]	+.0118 [+.0036, +.0200]
	BORROW-S	+.016 [+.005, +.028]	−.0002 [−.0013, +.0008]	−.0019 [−.0065, +.0026]
	CORRECTED-META	+.015 [−.003, +.033]	+.0003 [−.0017, +.0024]	+.0096 [+.0021, +.0172]
	GATE-BINARY	+.010 [−.008, +.027]	+.0007 [−.0009, +.0022]	+.0100 [+.0031, +.0169]
	GATE-BLOCK	+.008 [−.009, +.026]	+.0004 [−.0012, +.0019]	+.0091 [+.0027, +.0156]
	SUP	−.041 [−.072, −.010]	+.0045 [+.0007, +.0083]	+.0253 [+.0098, +.0408]
	CS	−.006 [−.032, +.021]	+.0026 [+.0001, +.0052]	+.0318 [+.0209, +.0426]
	SUPTRANS	+.004 [−.021, +.030]	+.0004 [−.0019, +.0026]	+.0090 [−.0002, +.0181]

Table S21: The same contrasts on VALID at $n_T = 400$. Layout and conventions are as in Table S20.

Branch	Endpoint	ΔAUC	ΔBrier	ΔECE
BIOVIL	BORROW-X	+.022 [+.005, +.039]	−.0031 [−.0041, −.0021]	−.0042 [−.0112, +.0027]
	BORROW-S	−.001 [−.007, +.005]	−.0001 [−.0008, +.0005]	+.0018 [−.0036, +.0071]
	CORRECTED-META	+.024 [+.006, +.041]	−.0035 [−.0044, −.0025]	−.0002 [−.0070, +.0066]
	GATE-BINARY	+.024 [+.006, +.041]	−.0035 [−.0044, −.0025]	−.0002 [−.0070, +.0066]
	GATE-BLOCK	+.019 [+.002, +.035]	−.0030 [−.0039, −.0020]	−.0002 [−.0070, +.0066]
	SUP	−.008 [−.022, +.005]	+.0016 [+.0004, +.0029]	+.0003 [−.0071, +.0076]
	CS	+.021 [+.005, +.037]	−.0023 [−.0033, −.0014]	−.0010 [−.0086, +.0066]
	SUPTRANS	+.012 [−.004, +.028]	−.0018 [−.0031, −.0006]	−.0077 [−.0147, −.0007]
SIMCLR	BORROW-X	−.023 [−.045, −.001]	+.0006 [−.0010, +.0022]	+.0067 [−.0002, +.0136]
	BORROW-S	+.013 [−.001, +.026]	−.0011 [−.0016, −.0007]	+.0040 [−.0027, +.0107]
	CORRECTED-META	−.008 [−.031, +.014]	−.0004 [−.0020, +.0013]	+.0064 [−.0006, +.0134]
	GATE-BINARY	−.005 [−.026, +.016]	−.0001 [−.0013, +.0010]	+.0043 [−.0012, +.0099]
	GATE-BLOCK	+.004 [−.014, +.022]	−.0007 [−.0018, +.0004]	+.0063 [−.0014, +.0139]
	SUP	−.056 [−.087, −.025]	+.0143 [+.0103, +.0184]	+.0577 [+.0466, +.0687]
	CS	−.031 [−.061, −.001]	+.0054 [+.0022, +.0085]	+.0476 [+.0416, +.0536]
	SUPTRANS	−.016 [−.047, +.014]	+.0005 [−.0006, +.0016]	+.0083 [−.0010, +.0175]
SUPERVISED	BORROW-X	+.012 [−.009, +.032]	+.0002 [−.0023, +.0026]	+.0101 [−.0008, +.0210]
	BORROW-S	+.008 [−.000, +.015]	−.0003 [−.0009, +.0003]	+.0009 [−.0043, +.0062]
	CORRECTED-META	+.019 [+.001, +.038]	−.0003 [−.0026, +.0020]	+.0105 [+.0005, +.0204]
	GATE-BINARY	+.010 [−.002, +.022]	−.0000 [−.0019, +.0018]	+.0091 [+.0014, +.0169]
	GATE-BLOCK	+.012 [+.000, +.025]	−.0002 [−.0021, +.0016]	+.0092 [+.0008, +.0176]
	SUP	−.030 [−.050, −.010]	+.0079 [+.0029, +.0128]	+.0275 [+.0121, +.0429]
	CS	−.007 [−.035, +.021]	+.0053 [+.0015, +.0090]	+.0339 [+.0208, +.0471]
	SUPTRANS	+.007 [−.016, +.030]	+.0012 [−.0015, +.0038]	+.0119 [+.0013, +.0225]

Table S22: Paired PR-AUC contrasts against TARGET-ONLY on VALID. The cohort event rate is 0.105, at which AUC is insensitive to the positive class, so PR-AUC is reported alongside it. It carries no part of the three-metric argument of Section 2.9, which concerns calibration. Layout and conventions are as in Table 15.

Endpoint	BIOVIL	SIMCLR	SUPERVISED
Panel A: $n_T = 200$			
BORROW-X	+.016 [−.006, +.039]	−.031 [−.075, +.014]	−.004 [−.026, +.018]
BORROW-S	+.022 [+.004, +.041]	+.041 [+.017, +.065]	+.003 [−.008, +.014]
CORRECTED-META	+.035 [+.012, +.058]	−.001 [−.043, +.041]	+.006 [−.020, +.033]
GATE-BINARY	+.035 [+.015, +.056]	+.012 [−.019, +.043]	+.001 [−.024, +.026]
GATE-BLOCK	+.030 [+.010, +.050]	+.028 [+.001, +.055]	+.001 [−.024, +.026]
SUP	−.050 [−.085, −.016]	−.068 [−.102, −.033]	−.042 [−.070, −.013]
CS	+.026 [−.002, +.054]	−.021 [−.061, +.019]	−.006 [−.034, +.022]
SUPTRANS	+.028 [+.008, +.047]	−.026 [−.069, +.016]	−.006 [−.032, +.021]
Panel B: $n_T = 400$			
BORROW-X	+.030 [+.009, +.052]	−.017 [−.048, +.015]	+.015 [−.012, +.042]
BORROW-S	+.006 [−.006, +.017]	+.015 [+.000, +.030]	+.003 [−.005, +.012]
CORRECTED-META	+.032 [+.012, +.053]	−.001 [−.034, +.033]	+.022 [−.006, +.050]
GATE-BINARY	+.032 [+.012, +.053]	−.004 [−.023, +.015]	+.012 [−.010, +.034]
GATE-BLOCK	+.028 [+.008, +.049]	+.005 [−.014, +.024]	+.014 [−.008, +.037]
SUP	−.039 [−.064, −.015]	−.055 [−.089, −.022]	−.021 [−.055, +.014]
CS	+.015 [−.011, +.042]	−.021 [−.063, +.021]	−.000 [−.038, +.037]
SUPTRANS	+.012 [−.015, +.038]	−.027 [−.060, +.006]	+.007 [−.025, +.038]

Tables S23 and S24 report endpoint levels on VALID. They situate the paired contrasts and carry no claim of their own. CEILING fits on labels every other row treats as absent, and CNN reads pixels; both are references rather than competitors.

Table S23: Endpoint levels on VALID at $n_T = 200$, as means over 20 repeats.

Branch	Endpoint	AUC	Brier	ECE
BIOVIL	TARGET-ONLY	.721	.0908	.0406
	BORROW-X	.743	.0874	.0282
	BORROW-S	.737	.0895	.0396
	CORRECTED-META	.754	.0863	.0308
	GATE-BINARY	.753	.0867	.0330
	GATE-BLOCK	.753	.0868	.0324
	SUP	.671	.0932	.0450
	CS	.751	.0875	.0378
	SUPTRANS	.744	.0868	.0283
	CEILING	.755	.0854	.0354
SIMCLR	TARGET-ONLY	.654	.0932	.0357
	BORROW-X	.653	.0933	.0434
	BORROW-S	.697	.0910	.0360
	CORRECTED-META	.686	.0910	.0411
	GATE-BINARY	.687	.0914	.0420
	GATE-BLOCK	.691	.0914	.0360
	SUP	.599	.1030	.0779
	CS	.660	.0987	.0783
	SUPTRANS	.650	.0920	.0240
	CEILING	.757	.0833	.0350
SUPERVISED	TARGET-ONLY	.667	.0953	.0486
	BORROW-X	.671	.0961	.0604
	BORROW-S	.684	.0950	.0467
	CORRECTED-META	.682	.0956	.0582
	GATE-BINARY	.677	.0959	.0586
	GATE-BLOCK	.676	.0956	.0577
	SUP	.626	.0998	.0739
	CS	.662	.0979	.0804
	SUPTRANS	.672	.0956	.0576
	CEILING	.729	.0868	.0370
CNN	—	.650	.0960	.0531

Table S24: Endpoint levels on VALID at $n_T = 400$, as means over 20 repeats. Layout and conventions are as in Table S23.

Branch	Endpoint	AUC	Brier	ECE
BIOVIL	TARGET-ONLY	.735	.0887	.0365
	BORROW-X	.757	.0855	.0322
	BORROW-S	.734	.0885	.0382
	CORRECTED-META	.758	.0852	.0363
	GATE-BINARY	.758	.0852	.0363
	GATE-BLOCK	.753	.0857	.0363
	SUP	.726	.0903	.0367
	CS	.756	.0863	.0354
	SUPTRANS	.747	.0868	.0287
	CEILING	.763	.0844	.0359
SIMCLR	TARGET-ONLY	.692	.0899	.0230
	BORROW-X	.670	.0905	.0297
	BORROW-S	.705	.0887	.0270
	CORRECTED-META	.684	.0895	.0294
	GATE-BINARY	.687	.0898	.0273
	GATE-BLOCK	.696	.0892	.0293
	SUP	.637	.1042	.0807
	CS	.661	.0953	.0706
	SUPTRANS	.676	.0904	.0313
	CEILING	.746	.0852	.0351
SUPERVISED	TARGET-ONLY	.692	.0916	.0494
	BORROW-X	.704	.0918	.0596
	BORROW-S	.700	.0914	.0504
	CORRECTED-META	.711	.0914	.0599
	GATE-BINARY	.702	.0916	.0585
	GATE-BLOCK	.705	.0914	.0586
	SUP	.662	.0995	.0769
	CS	.685	.0969	.0834
	SUPTRANS	.699	.0928	.0613
	CEILING	.750	.0848	.0324
CNN	—	.681	.0918	.0507

The same contrasts on TU are reported separately in Tables S25 and S26, because the representation, standardisation and imputation have read these rows. CEILING is in-sample on TU and is not scored.

Table S25: Paired contrasts against TARGET-ONLY on the transductive TU evaluation at $n_T = 200$. Layout and conventions are as in Table S20.

Branch	Endpoint	ΔAUC	ΔBrier	ΔECE
BIOVIL	BORROW-X	+.020 [+.006, +.034]	−.0037 [−.0044, −.0030]	−.0125 [−.0197, −.0052]
	BORROW-S	+.023 [+.006, +.040]	−.0015 [−.0027, −.0003]	+.0016 [−.0029, +.0060]
	CORRECTED-META	+.035 [+.020, +.051]	−.0052 [−.0061, −.0044]	−.0122 [−.0200, −.0044]
	GATE-BINARY	+.034 [+.019, +.050]	−.0049 [−.0059, −.0038]	−.0108 [−.0183, −.0034]
	GATE-BLOCK	+.032 [+.015, +.048]	−.0047 [−.0059, −.0035]	−.0115 [−.0186, −.0044]
	SUP	−.042 [−.061, −.022]	+.0018 [+.0005, +.0031]	−.0048 [−.0146, +.0050]
	CS	+.026 [+.008, +.043]	−.0044 [−.0053, −.0035]	−.0085 [−.0153, −.0016]
	SUPTRANS	+.013 [−.002, +.029]	−.0034 [−.0046, −.0022]	−.0154 [−.0249, −.0059]
SIMCLR	BORROW-X	−.023 [−.050, +.003]	+.0006 [−.0015, +.0026]	−.0069 [−.0152, +.0014]
	BORROW-S	+.044 [+.022, +.066]	−.0024 [−.0034, −.0013]	−.0001 [−.0036, +.0034]
	CORRECTED-META	+.012 [−.013, +.037]	−.0020 [−.0032, −.0008]	−.0138 [−.0204, −.0071]
	GATE-BINARY	+.015 [−.008, +.038]	−.0014 [−.0026, −.0002]	−.0081 [−.0142, −.0020]
	GATE-BLOCK	+.028 [+.012, +.044]	−.0019 [−.0029, −.0010]	−.0041 [−.0077, −.0004]
	SUP	−.085 [−.117, −.053]	+.0119 [+.0097, +.0140]	+.0359 [+.0311, +.0407]
	CS	−.028 [−.056, +.000]	+.0058 [+.0047, +.0068]	+.0291 [+.0240, +.0343]
	SUPTRANS	−.046 [−.076, −.016]	+.0002 [−.0008, +.0012]	−.0186 [−.0268, −.0104]
SUPERVISED	BORROW-X	+.006 [−.012, +.023]	−.0007 [−.0022, +.0007]	−.0018 [−.0089, +.0053]
	BORROW-S	+.017 [+.008, +.026]	−.0008 [−.0016, −.0001]	−.0006 [−.0029, +.0018]
	CORRECTED-META	+.021 [+.006, +.036]	−.0017 [−.0029, −.0004]	−.0039 [−.0111, +.0033]
	GATE-BINARY	+.020 [+.005, +.034]	−.0016 [−.0028, −.0004]	−.0043 [−.0112, +.0026]
	GATE-BLOCK	+.019 [+.005, +.033]	−.0017 [−.0028, −.0005]	−.0046 [−.0115, +.0022]
	SUP	−.048 [−.069, −.028]	+.0051 [+.0022, +.0080]	+.0196 [+.0101, +.0292]
	CS	−.024 [−.040, −.009]	+.0049 [+.0033, +.0065]	+.0287 [+.0220, +.0355]
	SUPTRANS	−.015 [−.036, +.006]	+.0017 [+.0001, +.0032]	+.0028 [−.0043, +.0100]

Table S26: The same contrasts on the transductive TU evaluation at $n_T = 400$.

Branch	Endpoint	ΔAUC	ΔBrier	ΔECE
BIOVIL	BORROW-X	+.019 [+.008, +.031]	−.0031 [−.0038, −.0024]	−.0067 [−.0107, −.0027]
	BORROW-S	+.007 [−.001, +.015]	−.0005 [−.0013, +.0003]	+.0018 [−.0011, +.0047]
	CORRECTED-META	+.025 [+.015, +.036]	−.0038 [−.0046, −.0031]	−.0072 [−.0109, −.0035]
	GATE-BINARY	+.025 [+.015, +.036]	−.0038 [−.0046, −.0031]	−.0072 [−.0109, −.0035]
	GATE-BLOCK	+.023 [+.012, +.035]	−.0036 [−.0045, −.0027]	−.0068 [−.0105, −.0030]
	SUP	−.025 [−.036, −.013]	+.0023 [+.0010, +.0036]	−.0009 [−.0076, +.0058]
	CS	+.013 [+.002, +.024]	−.0027 [−.0034, −.0020]	−.0102 [−.0141, −.0063]
	SUPTRANS	+.004 [−.007, +.015]	−.0015 [−.0025, −.0004]	−.0082 [−.0139, −.0025]
SIMCLR	BORROW-X	−.033 [−.044, −.022]	+.0008 [+.0003, +.0013]	−.0104 [−.0159, −.0049]
	BORROW-S	+.030 [+.020, +.039]	−.0021 [−.0025, −.0017]	+.0014 [−.0011, +.0039]
	CORRECTED-META	−.006 [−.019, +.006]	−.0010 [−.0015, −.0005]	−.0126 [−.0171, −.0081]
	GATE-BINARY	−.008 [−.019, +.002]	−.0004 [−.0007, −.0001]	−.0077 [−.0127, −.0026]
	GATE-BLOCK	+.010 [−.006, +.025]	−.0013 [−.0019, −.0007]	−.0027 [−.0069, +.0015]
	SUP	−.083 [−.103, −.063]	+.0191 [+.0172, +.0210]	+.0556 [+.0517, +.0595]
	CS	−.042 [−.058, −.027]	+.0069 [+.0058, +.0080]	+.0333 [+.0296, +.0371]
	SUPTRANS	−.054 [−.072, −.036]	+.0022 [+.0014, +.0030]	−.0058 [−.0111, −.0006]
SUPERVISED	BORROW-X	−.001 [−.013, +.011]	+.0022 [−.0000, +.0044]	+.0114 [+.0011, +.0216]
	BORROW-S	+.010 [+.002, +.018]	−.0009 [−.0017, −.0001]	−.0009 [−.0037, +.0019]
	CORRECTED-META	+.007 [−.004, +.018]	+.0006 [−.0009, +.0021]	+.0080 [−.0004, +.0164]
	GATE-BINARY	+.005 [−.005, +.014]	+.0004 [−.0007, +.0015]	+.0065 [−.0011, +.0141]
	GATE-BLOCK	+.007 [−.003, +.017]	+.0001 [−.0011, +.0013]	+.0055 [−.0025, +.0134]
	SUP	−.044 [−.059, −.029]	+.0094 [+.0061, +.0127]	+.0325 [+.0217, +.0433]
	CS	−.025 [−.038, −.011]	+.0060 [+.0040, +.0081]	+.0342 [+.0263, +.0421]
	SUPTRANS	−.013 [−.026, +.000]	+.0036 [+.0015, +.0057]	+.0129 [+.0048, +.0209]

S2.10 Negative-control analysis

Each permuted run reuses the split and label-free nuisance quantities of its real-source pair. Source labels are permuted within SL, preserving their marginal count while breaking their association with Z.

Paired gate decisions. Table S27 retains the pairing between the real and permuted runs rather than comparing acceptance rates alone.

Table S27: Real and permuted-source decisions at the applied ϵ_0. Counts are out of 20 paired repeats. The interval is the Clopper–Pearson 95% interval for the permuted acceptance rate. Discordance is real-only : permuted-only, and p is the exact two-sided McNemar test. Block acceptance means $\hat{c} \neq \varnothing$.

n_T	Branch	Rule	Real	Permuted	Permuted rate 95% CI	Discordance	p
200	BIOVIL	Binary	18	2	[.012, .317]	16 : 0	3.1×10^{-5}
		Block	18	2	[.012, .317]	16 : 0	3.1×10^{-5}
	SIMCLR	Binary	14	4	[.057, .437]	10 : 0	.0020
		Block	18	4	[.057, .437]	14 : 0	1.2×10^{-4}
400	BIOVIL	Binary	20	3	[.032, .379]	17 : 0	1.5×10^{-5}
		Block	20	4	[.057, .437]	16 : 0	3.1×10^{-5}
	SIMCLR	Binary	12	3	[.032, .379]	9 : 0	.0039
		Block	17	4	[.057, .437]	13 : 0	2.4×10^{-4}

Held-out performance after permutation. Table S28 scores the gated estimators on VALID. Agreeing refusals contribute zero to these contrasts.

Table S28: Paired contrasts against TARGET-ONLY on VALID after permuting the source labels. Entries are means with paired 95% t-intervals over 20 repeats. Positive ΔAUC and negative ΔBrier or ΔECE are improvements. The count records repeats with strictly positive ΔAUC.

n_T	Branch	Endpoint	ΔAUC	ΔAUC > 0	ΔBrier	ΔECE
200	BIOVIL	GATE-BINARY	$-.0011$ [$-.0031, +.0008$]	0/20	$-.00012$ [$-.00045, +.00021$]	$-.0025$ [$-.0076, +.0026$]
		GATE-BLOCK	$-.0011$ [$-.0031, +.0008$]	0/20	$-.00012$ [$-.00045, +.00021$]	$-.0025$ [$-.0076, +.0026$]
	SIMCLR	GATE-BINARY	$-.0051$ [$-.0158, +.0056$]	1/20	$+.00014$ [$-.00002, +.00031$]	$-.0007$ [$-.0022, +.0009$]
		GATE-BLOCK	$-.0051$ [$-.0158, +.0056$]	1/20	$+.00014$ [$-.00002, +.00031$]	$-.0007$ [$-.0022, +.0009$]
400	BIOVIL	GATE-BINARY	$+.0006$ [$-.0008, +.0020$]	1/20	$-.00013$ [$-.00030, +.00003$]	$-.0006$ [$-.0013, +.0002$]
		GATE-BLOCK	$+.0014$ [$-.0008, +.0036$]	2/20	$-.00028$ [$-.00062, +.00007$]	$-.0003$ [$-.0024, +.0018$]
	SIMCLR	GATE-BINARY	$-.0112$ [$-.0303, +.0079$]	1/20	$+.00057$ [$-.00039, +.00154$]	$+.0011$ [$-.0013, +.0035$]
		GATE-BLOCK	$-.0099$ [$-.0293, +.0096$]	2/20	$+.00049$ [$-.00050, +.00149$]	$+.0012$ [$-.0012, +.0036$]

Null calibration of the threshold. The control is reported rather than used to select ϵ_0. Table S29 gives the smallest grid value at which the permuted acceptance rate is at most .10. With 20 repeats the rate moves in increments of .05, so these values locate a grid crossing, not a calibrated constant.

Table S29: Applied and null-calibrated thresholds. The applied value is $\log(d)/n_T$. The last two columns give the smallest grid threshold at which the corresponding permuted acceptance rate is at most .10.

			Permuted rate at applied ϵ_0		Null-calibrated ϵ_0	
n_T	Branch	Applied ϵ_0	Binary	Block	Binary	Block
200	BIOVIL	.0243	.10	.10	.010	.020
	SIMCLR	.0270	.20	.20	.050	.050
400	BIOVIL	.0122	.15	.20	.030	.030
	SIMCLR	.0135	.15	.20	.020	.020

The applied and null-calibrated values differ in both directions and by as much as a factor of 2.5. Choosing the applied value from this diagnostic would make the acceptance rates in Table S27 partly a consequence of the choice.

Threshold replay. Table S30 replays the paired separation without refitting. The applied row uses the branch-specific values above. Entries greater than .05 mark thresholds at which the corresponding rule no longer separates the real and permuted sources by the exact McNemar test.

Table S30: Exact two-sided McNemar p-values for real-versus-permuted acceptance across the threshold grid. B and K denote GATE-BINARY and GATE-BLOCK.

(a) $n_T = 200$

	BIOVIL		SIMCLR	
ϵ_0	B	K	B	K
.000	.0020	.0625	.0225	.0625
.005	$< .0001$	.0005	.0063	.0078
.010	$< .0001$	$< .0001$	.0063	.0005
.015	$< .0001$	$< .0001$	.0005	.0001
.020	$< .0001$	$< .0001$	.0020	$< .0001$
applied	$< .0001$	$< .0001$	.0020	.0001
.030	$< .0001$	$< .0001$	.0039	.0002
.040	$< .0001$	$< .0001$	.0625	.0312
.050	.0156	.0078	.0312	.0312
.075	.2500	.2500	1.000	.6250

(b) $n_T = 400$

	BIOVIL		SIMCLR	
ϵ_0	B	K	B	K
.0000	.0020	.5000	.0312	.0312
.0010	.0005	.0312	.0312	.0156
.0020	.0002	.0078	.0391	.0117
.0030	.0001	.0078	.1250	.0117
.0050	$< .0001$	.0010	.1250	.0117
applied	$< .0001$	$< .0001$	.0039	.0002
.0150	$< .0001$	$< .0001$	.0078	.0001
.0200	$< .0001$	$< .0001$	.0039	.0002
.0300	.0002	.0002	.0078	.0039
.0500	.0156	.0156	.2500	.2500

At $n_T = 400$, the binary SIMCLR comparison lapses over $\epsilon_0 \in [.003, .005]$ and returns before the applied value; the block comparison remains below .05 throughout that range. The non-monotonicity comes from changes in the paired discordances, not from refitting the estimators.